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# Measurement Without Collapse: Phase Constraint Resolution

Phase Theory VI

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### Abstract

The measurement problem in quantum mechanics — the question of why measurement produces definite outcomes from superposed quantum states, and why this process appears to require a non-unitary collapse distinct from the Schrödinger equation — represents the longest-standing conceptual difficulty in the foundations of physics. Since von Neumann's formal articulation in 1932, every proposed resolution has either introduced additional ontological machinery (collapse parameters, many branches, pilot waves, agent beliefs) or retreated into operational agnosticism. This paper demonstrates that within Phase Theory, the measurement problem is not solved but dissolved: it is shown to be an artifact of misidentified ontological primitives rather than a genuine feature of physical reality. Within Phase Theory, physical reality consists entirely of a phase-bearing substrate whose global organization is governed by the admissibility functional $\mathscr{I}[\Phi] \geq 0$. Measurement is shown to be an ordinary physical interaction in which a system phase configuration $\Phi_s$ couples to an apparatus phase configuration $\Phi_a$ within an environmental context $\Phi_e$, forming a composite $\Phi_{total} = \Phi_s \oplus \Phi_a \oplus \Phi_e$ whose evolution is governed entirely and deterministically by global admissibility. Discrete outcomes arise without collapse from the topological structure of surviving admissible configurations (winding numbers); irreversibility arises without a fundamental arrow of time from combinatorial admissibility suppression; Born-rule statistics arise without probabilistic postulates from the admissibility measure on phase configuration space (derived in Paper 5 of this series); observer privilege is eliminated by treating observers as macroscopic phase structures subject to the same admissibility rules as all other systems. The paper provides formal proofs of five central results: (Theorem 6.1) Phase Theory admits no measurement postulate; (Theorem 6.2) the Born rule is derived from admissibility measure; (Theorem 6.3) Bell correlations are reproduced by global admissibility without hidden variables; (Theorem 6.4) classicality is the high-constraint-density limit of admissibility dynamics; and (Proposition 6.1) measurement outcomes are topologically stable admissible resolutions. The theory is shown to be consistent with all known measurement phenomena, to reproduce all quantum mechanical predictions in tested regimes, and to yield six classes of novel, falsifiable predictions distinguishing Phase Theory from both standard quantum mechanics and objective collapse theories. The significance of these results is that, for the first time, a coherent and complete ontological framework is provided in which measurement requires no special postulate, no preferred basis selection mechanism, and no observer — the measurement problem disappears because the framework that generated it no longer applies.

## CONTENTS

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## 1. Introduction — The Measurement Problem in Historical Context

The measurement problem is the central unsolved conceptual difficulty of quantum mechanics. Its persistence across nearly a century of theoretical development — during which quantum mechanics itself has achieved unparalleled empirical success — testifies to the depth of the conceptual difficulty rather than any deficiency in the theory's predictive power. The problem arises from a fundamental bifurcation within the quantum mechanical formalism: on the one hand, the Schrödinger equation describes the evolution of quantum states as continuous, linear, and unitary; on the other hand, the measurement postulate asserts that upon measurement, the quantum state undergoes an instantaneous, non-linear, and irreversible collapse to an eigenstate of the measured observable. These two dynamical rules are formally incompatible, and no derivation of one from the other has ever been achieved within the standard framework.

John von Neumann, in his landmark 1932 treatise *Mathematische Grundlagen der Quantenmechanik* [1], gave the first rigorous formulation of this bifurcation. He distinguished two types of evolution — Process 1 (collapse, associated with measurement) and Process 2 (unitary evolution, associated with dynamics) — and noted that no criterion internal to the formalism determines when Process 1 applies. He introduced the concept of the "von Neumann chain": a sequence of quantum systems, each measuring the next, terminating in the observer's consciousness, with no objective location for the cut between the quantum and classical descriptions. This chain remains one of the clearest formulations of the problem's depth, because it shows that the difficulty is not merely technical but conceptual: the theory does not contain within itself the resources to determine where quantum ends and classical begins.

The Copenhagen interpretation, developed principally by Bohr and Heisenberg [2, 3, 22], responded to this difficulty by embracing instrumentalism: quantum mechanics is a theory of preparation and measurement outcomes, and it is meaningless to ask about the quantum state of a system independently of a specified experimental arrangement. Bohr's doctrine of complementarity — that experimental arrangements that reveal particle-like and wave-like properties are mutually exclusive and cannot be combined — articulates this instrumentalism at the level of experimental practice. While operationally successful, the Copenhagen interpretation provides no ontological account of what happens during a measurement. It dissolves the problem by refusing to engage with it, a strategy that satisfied working physicists but left foundationalists — and Einstein above all — profoundly dissatisfied [4].

Hugh Everett's 1957 relative-state formulation [5] proposed the most radical resolution: the Schrödinger equation is always correct and applies universally, including to apparatus and observer. Measurement does not produce collapse; it produces branching. Both outcomes of a measurement exist simultaneously in distinct branches of a proliferating universal wavefunction. The "many worlds" interpretation, developed from Everett's proposal by DeWitt and subsequently by Wallace [6, 49], attempts to make this precise. It faces, however, severe difficulties: the preferred basis problem (what determines the branches?), the probability problem (what grounds the Born rule in a deterministic branching picture?), and the ontological extravagance of positing a vast multiplicity of simultaneously existing realities.

David Bohm's 1952 hidden-variable theory [11] restores determinism and realism by positing that particles have definite positions at all times, guided by a pilot wave satisfying a modified Schrödinger equation. Measurement outcomes are determined by the particle's actual position (a hidden variable) and the guiding wave. Bohmian mechanics is empirically equivalent to standard quantum mechanics in all tested regimes and is internally consistent, but it introduces a non-local, instantaneous pilot wave with no independent physical motivation, and it extends awkwardly to relativistic and field-theoretic contexts.

Ghirardi, Rimini, and Weber [7], followed by Pearle's continuous spontaneous localization model [38], proposed objective collapse theories: the Schrödinger equation is modified by stochastic non-linear terms that localize wavefunctions in space. These theories make genuinely different empirical predictions from standard quantum mechanics and are thus testable in principle, but they introduce free parameters (collapse rate $\lambda$ and localization width $r_C$ in GRW) whose values must be chosen by hand, and they face serious difficulties with relativistic covariance and energy conservation.

The decoherence program, initiated by Zeh [9] and developed extensively by Zurek [8, 46], Joos et al. [10], and Schlosshauer [39, 40], demonstrates that quantum systems interacting with macroscopic environments rapidly lose the ability to display interference between measurement outcomes. Environmental entanglement suppresses off-diagonal terms in the density matrix, yielding an effective classical probability distribution over outcomes. Decoherence is an important physical phenomenon and explains why quantum superpositions are not directly observed in macroscopic systems, but it does not solve the measurement problem: decoherence explains why *we cannot observe* the coherence between outcomes; it does not explain why there *is* a definite outcome at all. The mixed state produced by decoherence is an improper mixture that, within unitary quantum mechanics, remains a superposition.

QBism — quantum Bayesianism, developed by Fuchs, Mermin, and Schack [13] — interprets quantum states as agents' degrees of belief rather than objective physical states. Measurement outcomes are the experiences of individual agents, and probabilities are subjective. QBism is internally consistent but radically anti-realist: it provides no account of the physical world independent of agents' beliefs, and the question of what the world is actually doing is declared meaningless.

The foregoing survey — necessarily compressed but representative — reveals that after nearly a century, the measurement problem remains unresolved. Every major approach either introduces new ontological machinery, invokes some form of observer privilege, abandons realism, or makes unverified empirical predictions. The present paper argues that this persistent failure reflects a more fundamental error: the assumption that space, time, particles, fields, and the wavefunction are ontological primitives, when in fact they are emergent structures from a more fundamental phase-theoretic substrate. Once this misidentification is corrected — as Phase Theory accomplishes — the measurement problem does not require a solution. It disappears.

The remainder of this paper is organized as follows. Section 2 reviews the Phase Theory foundations necessary for the present analysis. Section 3 reformulates the measurement problem within phase ontology and demonstrates that its four pillars cannot be stated therein. Section 4 proves that no measurement postulate is needed. Sections 5–10 develop the formal theory of measurement as phase constraint resolution, including the mechanism of outcome selection, the topology of discrete outcomes, and the derivation of irreversibility. Sections 11–19 apply the framework to decoherence, the Born rule, observer effects, contextuality, Wigner's friend, delayed choice, Bell correlations, and quantum erasure. Sections 20–23 compare Phase Theory with competitor frameworks. Sections 24–29 address experimental consistency, novel predictions, the quantum-classical divide, quantum information, philosophical implications, and gravitational decoherence. Section 30 concludes.

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## 2. Phase Theory — Foundations Relevant to Measurement

Phase Theory, developed in the preceding five papers of this series [41–45], proposes that phase is the sole fundamental constituent of physical reality. No background spacetime, no primitive particles, no quantum fields, and no probability postulates are assumed. Instead, physical reality consists of a phase-bearing substrate $\Phi$ — a configuration in a topological phase space $\mathscr{M}_\Phi$ — whose physical admissibility is governed at every level by a single master functional, the **admissibility functional** $\mathscr{I}[\Phi]$. All physical laws, all symmetries, and all observable phenomena are to be understood as emergent features of configurations satisfying $\mathscr{I}[\Phi] \geq 0$.

The axiomatic structure of Phase Theory was established in Paper 1 [41]. The theory rests on four axioms. **Axiom I (Phase Primacy)**: Physical reality is constituted entirely by phase configurations $\Phi \in \mathscr{M}_\Phi$, where $\mathscr{M}_\Phi$ is a topological configuration space equipped with a natural measure. **Axiom II (Global Admissibility)**: A phase configuration $\Phi$ is physically realized if and only if $\mathscr{I}[\Phi] \geq 0$. **Axiom III (Topological Stability)**: Stable physical structures correspond to topologically stable sectors of $\mathscr{M}_\Phi$, characterized by discrete invariants (winding numbers, Chern classes). **Axiom IV (Compositional Closure)**: The composition of two phase configurations $\Phi_1$ and

$\Phi_2$ under boundary-compatible gluing is a well-defined phase configuration $\Phi_1 \oplus \Phi_2 \in \mathscr{M}_\Phi$.

Paper 2 [42] derived emergent spacetime geometry from phase curvature, establishing that Riemannian geometry and Einstein field equations appear as effective descriptions of the large-scale organization of slowly varying admissible phase configurations. Paper 3 [43] established topological discreteness: the particle spectrum of the Standard Model corresponds to distinct topological sectors of $\mathscr{M}_\Phi$, each characterized by a winding number vector $\mathbf{n} = (n_1, n_2, \ldots, n_k) \in \mathbb{Z}^k$. Mass, charge, and spin emerge from topological invariants of these sectors. Paper 4 [44] demonstrated that gauge interactions — electromagnetism, weak force, strong force — arise from phase topology: gauge bosons are phase excitations connecting topological sectors, and gauge symmetries are reparametrizations of phase configurations that preserve $\mathscr{I}[\Phi]$. Paper 5 [45] is most directly relevant to the present work: it established that probability is not a primitive but is the admissibility measure $\mu_\mathscr{I}$ on phase configuration space. Specifically, the probability of a physical event corresponds to the $\mu_\mathscr{I}$-measure of the set of phase configurations consistent with that event.

For the present paper, the most important objects are the following. The **phase composition** $\Phi_{total} = \Phi_s \oplus \Phi_a \oplus \Phi_e$ denotes the composite phase configuration of a system (s), apparatus (a), and environment (e), formed by topological gluing along shared boundaries. The **admissibility functional** $\mathscr{I}[\Phi_{total}] \geq 0$ is the master constraint on the composite. The **admissibility measure** $\mu_\mathscr{I}$ is the natural measure on the space of admissible configurations, derived in Paper 5 [45]. The key result of Paper 5 that we shall use in Section 13 is the Admissibility–Born Correspondence: in the coherent limit, $\mu_\mathscr{I}(\text{sector } i) = |\psi_i|^2$, where $\psi_i$ are the standard complex amplitudes of quantum mechanics. We also draw on Paper 3's topological discreteness: stable phase configurations are characterized by discrete winding numbers, and transitions between winding sectors require passing through configurations of strictly negative $\mathscr{I}$ — a process forbidden by the admissibility constraint.

In what follows, all prior results from Papers 1–5 are used without re-derivation but with specific citations to the relevant propositions and theorems in those papers.

## 3. The Measurement Problem Reframed Within Phase Ontology

The measurement problem in standard quantum mechanics rests on four distinct but interrelated pillars, each of which presupposes certain ontological primitives. We examine each pillar and show that, within phase ontology, none can be formulated in its standard form. This is not a technical limitation of Phase Theory but a conceptual virtue: the problem is dissolved rather than requiring resolution within its own terms.

**Pillar I: Continuous unitary dynamics.** Standard quantum mechanics assumes that the state of a system evolves according to the Schrödinger equation $i\hbar\, \partial|\psi\rangle/\partial t = \hat{H}|\psi\rangle$ between measurements. This evolution is continuous, linear, and unitary. Within phase ontology, there is no primitive Hilbert space $|\psi\rangle$, no background time $t$ relative to which to define $\partial/\partial t$, and no Hamiltonian $\hat{H}$ as a primitive operator. Instead, phase evolution is governed by admissibility: a phase trajectory $\Phi(\lambda)$ through $\mathscr{M}_\Phi$ is admissible if $\mathscr{I}[\Phi(\lambda)] \geq 0$ for all values of the parameter $\lambda$. This is a constraint, not an equation of motion, and it is global rather than local. The Schrödinger equation emerges in Paper 5 [45] as an effective description valid for coherent propagating phase modes — it is an approximation, not a fundamental law.

**Pillar II: Discontinuous collapse.** Upon measurement, the quantum state $|\psi\rangle = \Sigma_i\, c_i|i\rangle$ collapses to a single eigenstate $|k\rangle$ with probability $|c_k|^2$. This collapse is discontinuous, non-linear, and irreversible. Within phase ontology, there are no quantum states in Hilbert space and no eigenstate expansion. Phase configurations evolve by continuous admissibility constraint satisfaction. The apparent discontinuity of collapse is an artifact of describing a topological transition — from a superposition of winding sectors to a single stable winding sector — in the language of wavefunctions. The underlying phase transition is governed by continuous admissibility variation; see Section 8 for the detailed mechanism.

**Pillar III: Probabilistic outcomes.** Standard quantum mechanics predicts that measurement outcomes occur with probabilities given by the Born rule. These probabilities are irreducible: they cannot be explained by deeper deterministic dynamics. Within phase ontology, probability is the admissibility measure $\mu_\mathscr{I}$ (Paper

5 [45]). This measure is derived, not postulated, from the admissibility functional itself. It is therefore not an irreducible primitive but a consequence of the global phase-theoretic structure. The appearance of irreducible randomness is an artifact of ignoring the full global structure of $\mathscr{M}_\Phi$ and coarse-graining over it.

**Pillar IV: Observer privilege.** Many formulations of the measurement problem accord a special role to the observer: in Copenhagen, the observer's choice of measurement establishes which observable has a definite value; in von Neumann's chain, the observer's consciousness terminates the regress; in QBism, the observer's degrees of belief just are the quantum state. Within phase ontology, the observer is a macroscopic phase structure $\Phi_{obs} \in \mathscr{M}_\Phi$ governed by $\mathscr{I}[\Phi_{obs}] \geq 0$, in no way ontologically privileged relative to any other macroscopic phase configuration. Extending the composite to include the observer changes the admissibility constraints but not their character: $\mathscr{I}[\Phi_s \oplus \Phi_a \oplus \Phi_e \oplus \Phi_{obs}] \geq 0$. No new physics is introduced. Observer privilege is dissolved.

Since none of the four pillars can be stated in phase ontology — because the primitives they presuppose (Hilbert space, background time, irreducible probability, observer privilege) are not present — the measurement problem in its standard form cannot even be posed within Phase Theory. This is the sense in which Phase Theory dissolves rather than solves the measurement problem. We proceed to develop the positive theory: what measurement actually is, and why it produces the phenomena that motivated the measurement problem.

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## 4. No Primitive Measurement Postulate

We establish here the foundational negative result of this paper: Phase Theory admits no measurement postulate, and any attempt to introduce one would be logically redundant.

**Theorem 6.1 (No Measurement Postulate).**

> Within Phase Theory, no measurement postulate is required or admissible. More precisely: given Axioms I–IV of Phase Theory [41], together with the admissibility functional $\mathscr{I}[\Phi] \geq 0$, the topological discreteness of stable configurations (Paper 3 [43]), the emergence of probability from admissibility measure (Paper 5 [45]), and the non-primacy of observers (Section 14 below), there exists no logical space for an independent collapse postulate. Any collapse postulate added to Phase Theory would be either contradicted by existing axioms or derivable from them, and hence is unnecessary.

***Proof sketch.*** *We proceed by exhaustion of the functions that collapse serves in standard QM. Collapse serves three functions: (a) it selects a definite outcome from superposed possibilities; (b) it makes that outcome discrete rather than continuous; (c) it assigns outcome probabilities. We show each function is already performed by Phase Theory. For (a): definite outcome selection is performed by admissibility pruning — global constraint enforcement eliminates all but one topologically stable configuration (Section 8–9). For (b): discreteness is provided by topological winding numbers (Paper 3 [43]), which are discrete by definition, without any jump or collapse required. For (c): outcome probabilities are provided by the admissibility measure* $\mu\_\mathscr{I}$ *(Paper 5 [45]). Since all three functions of collapse are already performed by other, more fundamental mechanisms, no collapse postulate can be added that does not conflict with existing machinery (contradiction) or merely renames it (redundancy). In either case, the postulate is inadmissible.* ∎

Theorem 6.1 establishes a powerful structural point. The standard measurement postulate is not merely an awkward feature of quantum mechanics; it is a symptom of an ontological misidentification. When the correct primitives are identified — phase configurations, global admissibility, topological stability — collapse becomes not just unnecessary but impossible to coherently state. A theory that still requires collapse has, by Theorem 6.1, misidentified at least one of its ontological primitives: it is either not treating probability as emergent, not treating discreteness as topological, or not treating outcome selection as admissibility-governed. Theorem 6.1 thus serves as a diagnostic criterion for ontological adequacy in quantum foundational theories.

It is important to be precise about what Theorem 6.1 does not claim. It does not claim that there are no sudden-seeming transitions in phase configurations. There are, as Section 8 will show, rapid admissibility-driven transitions that can appear sudden on timescales relevant to macroscopic observation. The claim is that these transitions are continuous admissibility-constraint satisfaction processes, not postulated jumps, and they are fully deterministic given the global phase configuration. The appearance of suddenness is scale-dependent: what is rapid on the decoherence timescale $\tau_{dec}$ is continuous in phase configuration space.

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## 5. Formal Definition — Measurement as Phase Coupling

Having established the negative result of Section 4, we turn to the positive theory. We begin with the fundamental definition.

> **Definition 6.1 (Measurement).**
>
> A
>
> measurement interaction
>
> is a physical interaction in which a system phase configuration $\Phi_s \in \mathcal{M}_\Phi$ couples to an apparatus phase configuration $\Phi_a \in \mathcal{M}_\Phi$ within an environmental context $\Phi_e \in \mathcal{M}_\Phi$, forming the composite configuration
>
> (6.1) $\Phi_{total} = \Phi_s \oplus \Phi_a \oplus \Phi_e$
>
> subject to the global admissibility constraint $\mathcal{I}[\Phi_{total}] \geq 0$, where the apparatus satisfies the
>
> macroscopicity condition
>
> : its constraint density $\rho_c[\Phi_a] \geq \rho_*$ for a theory-determined threshold $\rho_*$, and where the system–apparatus coupling produces non-trivial admissibility correlations between the system's topological sector and the apparatus's pointer

> configuration.

Each component of Definition 6.1 requires careful elaboration. The system $\Phi_s$ is a phase configuration in a topological sector of $\mathscr{M}_\Phi$; in the coherent limit, this corresponds to a superposition of winding sectors. The apparatus $\Phi_a$ is a macroscopic phase configuration — meaning it encompasses O(N) coupled degrees of freedom for $N \sim 10^{23}$ in laboratory-scale experiments. This macroscopic character is not merely scale; it is the source of the apparatus's enormous constraint-generating power, which is formalized by the constraint density $\rho_c[\Phi_a]$ defined below. The environment $\Phi_e$ encompasses all remaining phase degrees of freedom that interact non-trivially with the composite $\Phi_s \oplus \Phi_a$; it is in principle unbounded but in practice characterized by a finite correlation length beyond which its admissibility constraints decouple.

The composition operator ⊕ in equation (6.1) is not simple addition of phase values; it encodes a topological gluing of phase configurations along shared boundaries in $\mathscr{M}_\Phi$, subject to boundary compatibility conditions imposed by $\mathscr{I}$. The detailed mathematics of ⊕ is developed in Section 6. The admissibility functional $\mathscr{I}[\Phi_{total}]$ acts on the full composite and cannot be decomposed into independent contributions from each component; this non-separability is essential and is the formal seat of entanglement (Section 6).

The **Constraint Density** $\rho_c[\Phi_a]$ is defined as follows. Let $C[\Phi_a]$ be the set of independent admissibility constraints that the apparatus phase configuration introduces into the composite system — that is, the set of constraints of the form $f_i(\Phi_{total}) \geq 0$ that are not satisfied by all of $\mathscr{M}_\Phi$ and whose satisfaction depends non-trivially on the apparatus sector. Then:

$$\rho_c[\Phi_a] := |C[\Phi_a]| \,/\, Vol_{\mathscr{M}}(supp\ \Phi_a) \tag{6.2}$$

where $Vol_{\mathscr{M}}$ denotes the natural volume measure on $\mathscr{M}_\Phi$ and supp $\Phi_a$ is the support of the apparatus configuration in $\mathscr{M}_\Phi$. For a laboratory apparatus, $|C[\Phi_a]| \sim N_a$ (the number of apparatus degrees of freedom, of order $10^{23}$), giving $\rho_c[\Phi_a]$

$\gg 1$ in natural units. The macroscopicity threshold $\rho_*$ corresponds to the constraint density above which the apparatus can robustly distinguish topological sectors of the system — that is, above which admissibility pruning reliably selects a unique outcome (Section 7).

Definition 6.1 distinguishes measurement from generic phase interactions by two criteria: macroscopicity ($\rho_c[\Phi_a] \geq \rho_*$) and non-trivial admissibility correlation between system topology and apparatus pointer configuration. Generic microscopic interactions between two phase configurations involve low constraint density and do not necessarily produce robust topological sector selection. Measurement is the special case where the apparatus is large enough that its constraints overwhelmingly determine which system topological sector survives the admissibility filter. This is not a distinction in kind — it is a distinction in degree that becomes sharp in the thermodynamic limit $N_a \to \infty$.

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## 6. The Composite Phase Space and Its Structure

The central mathematical object of this paper is the composite phase configuration $\Phi_{total} = \Phi_s \oplus \Phi_a \oplus \Phi_e$. We now develop the mathematics of this composition in detail, emphasizing properties that are crucial for understanding measurement.

Let $\mathscr{M}_\Phi^{(s)}$, $\mathscr{M}_\Phi^{(a)}$, $\mathscr{M}_\Phi^{(e)}$ denote the phase configuration spaces of the system, apparatus, and environment respectively. Each is equipped with a topology $T_s$, $T_a$, $T_e$ and a natural measure. The naive composite space would be the Cartesian product $\mathscr{M}_\Phi^{(s)} \times \mathscr{M}_\Phi^{(a)} \times \mathscr{M}_\Phi^{(e)}$, equipped with the product topology. However, the composition operation $\oplus$ in Phase Theory is not Cartesian product but topological gluing subject to boundary compatibility conditions. Formally, let $\partial\mathscr{M}_\Phi^{(s)}$ denote the boundary of the system phase space in $\mathscr{M}_\Phi$, and similarly for apparatus and environment. The gluing requires that:

$$\text{(6.3)} \quad \partial\mathscr{M}_\Phi^{(s)} \cap \partial\mathscr{M}_\Phi^{(a)} \neq \varnothing \quad \text{and} \quad \partial\mathscr{M}_\Phi^{(a)} \cap \partial\mathscr{M}_\Phi^{(e)} \neq \varnothing$$

with phase values matched continuously across these intersections according to the phase gluing conditions of Axiom IV [41]. The composite space $\mathscr{M}_\Phi^{(total)}$ =

$\mathscr{M}\_\Phi^{(s)} \oplus_\partial \mathscr{M}\_\Phi^{(a)} \oplus_\partial \mathscr{M}\_\Phi^{(e)}$ is a quotient of the product space under this gluing, and its topology is generically non-trivial even when each component space is topologically simple. This non-trivial topology of the composite is the Phase Theory counterpart of entanglement.

The admissibility functional on the composite is defined by:

$$(6.4)\quad \mathscr{I}[\Phi_{total}] = \int_{\mathscr{M}\_\Phi^{(total)}} \mathscr{A}(\Phi_{total}(x))\, d\mu(x)$$

where $\mathscr{A}$ is the local admissibility density (a smooth functional of the phase configuration and its derivatives up to order k) and dμ is the natural measure on $\mathscr{M}\_\Phi^{(total)}$. Crucially, the integration domain is the full composite space, not the product of component spaces, and $\mathscr{A}$ involves cross-terms between the system, apparatus, and environment sectors. This yields the fundamental non-separability result:

**Proposition 6.0 (Non-Separability of Admissibility).**

In general, $\mathscr{I}[\Phi_{total}] \neq \mathscr{I}[\Phi_s] + \mathscr{I}[\Phi_a] + \mathscr{I}[\Phi_e]$. Specifically, there exist cross-term contributions of the form:

$$(6.5)\quad \mathscr{I}[\Phi_{total}] = \mathscr{I}[\Phi_s] + \mathscr{I}[\Phi_a] + \mathscr{I}[\Phi_e] + \mathscr{I}\_{int}[\Phi_s, \Phi_a] + \mathscr{I}\_{int}[\Phi_a, \Phi_e] + \mathscr{I}\_{int}[\Phi_s, \Phi_a, \Phi_e]$$

where the interaction terms $\mathscr{I}\_{int}$ are generically non-zero whenever the composite topology is non-trivial. The non-separability of $\mathscr{I}$ means that admissibility constraints on the composite cannot be decomposed into independent constraints on each component; the admissibility of the whole is not determined by the admissibility of the parts. This is the formal Phase Theory source of entanglement: entangled quantum states correspond, in Phase Theory, to composite phase configurations whose admissibility is not separable into component admissibilities.

From equation (6.5), we derive the following structural result about the space of globally admissible composite configurations.

**Proposition 6.0b (Admissibility Restriction).**

Let $\mathscr{A}_s = \{\Phi_s : \mathscr{I}[\Phi_s] \geq 0\}$, and define $\mathscr{A}_a$, $\mathscr{A}_e$ analogously. Let $\mathscr{A}_{total} = \{\Phi_{total} : \mathscr{I}[\Phi_{total}] \geq 0\}$. Then:

$$\mathscr{A}_{total} \subsetneq \mathscr{A}_s \times \mathscr{A}_a \times \mathscr{A}_e \tag{6.6}$$

***Proof.*** *Let $\Phi_{total} \in \mathscr{A}_{total}$, so $\mathscr{I}[\Phi_{total}] \geq 0$. By equation (6.5), $\mathscr{I}[\Phi_s] + \mathscr{I}[\Phi_a] + \mathscr{I}[\Phi_e] + \mathscr{I}_{int} \geq 0$. This implies $\mathscr{I}[\Phi_s] \geq -\mathscr{I}_{int} - \mathscr{I}[\Phi_a] - \mathscr{I}[\Phi_e]$, which does not in general imply $\mathscr{I}[\Phi_s] \geq 0$. Hence $\mathscr{A}_{total} \subseteq \mathscr{A}_s \times \mathscr{A}_a \times \mathscr{A}_e$. Strictness follows from the existence of configurations in $\mathscr{A}_s \times \mathscr{A}_a \times \mathscr{A}_e$ for which $\mathscr{I}_{int} < -(\mathscr{I}[\Phi_s] + \mathscr{I}[\Phi_a] + \mathscr{I}[\Phi_e])$, violating $\mathscr{I}[\Phi_{total}] \geq 0$. Such configurations exist whenever the composite topology is non-trivial and the coupling is non-zero.* ∎

Equation (6.6) establishes that globally admissible composite configurations form a proper subset of the product of individually admissible component configurations. This is the mathematical statement that entanglement restricts the allowed joint configurations beyond what is imposed by the individual constraints. The measurement problem, in standard QM terms, asks why only one term in the tensor product expansion survives after measurement. Proposition 6.0b answers this: the globally admissible subset of the composite restricts to a much smaller set than the product of component admissible sets; for macroscopic apparatus, this restriction is so severe that it selects a unique topological sector (Section 7).

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## 7. Constraint Proliferation from Macroscopic Apparatus

The apparatus is distinguished from a generic microscopic phase configuration by its macroscopic character: it introduces $O(N_a)$ independent admissibility constraints on the composite, where $N_a \sim 10^{23}$ for a laboratory apparatus. This section quantifies the consequences of this constraint proliferation for the space of admissible composite configurations.

Let $C_i(\Phi_{total}) \geq 0$, $i = 1, \ldots, N_c$ denote the set of independent admissibility constraints that the apparatus introduces into the composite. Here $N_c = |C[\Phi_a]| \approx \alpha N_a$ for some O(1) constant $\alpha$ that depends on the structure of the admissibility functional for the specific apparatus type. Each constraint $C_i$ eliminates from the composite configuration space a subset of $\Phi_{total}$ configurations. Assuming — as is generic for a macroscopic apparatus in thermal contact with the environment — that these constraints are approximately independent, the fraction of composite configurations surviving all $N_c$ constraints is bounded above by:

$$f_{admissible} \leq \prod_{i=1}^{N_c} (1 - p_i) \tag{7.1}$$

where $p_i \in (0,1)$ is the fraction of configurations eliminated by constraint $C_i$. For generic apparatus with $p_i \geq p_* > 0$ for all i, this gives $f_{admissible} \leq (1 - p_*)^{N_c}$. Taking logarithms:

$$\ln f_{admissible} \leq N_c \ln(1 - p_*) \approx -N_c\, p_* = -\alpha\, p_*\, N_a \tag{7.2}$$

Hence $f_{admissible} \leq \exp(-\alpha\, p_*\, N_a)$. Setting $\alpha' = \alpha p_* > 0$, we obtain the admissibility suppression bound:

$$f_{admissible} \leq \exp(-\alpha' N_a) \tag{7.3}$$

For $N_a \sim 10^{23}$, equation (7.3) gives $f_{admissible} \leq \exp(-\alpha' \times 10^{23})$ — a number so fantastically small that the surviving configurations are, for all practical and theoretical purposes, a set of measure zero within the full composite configuration space. This is the mechanism by which macroscopic apparatus dramatically narrows the space of admissible outcomes: not by any discontinuous jump, but by the relentless application of an astronomical number of independent constraints.

**Lemma 6.1 (Admissibility Localization).**

For a macroscopic apparatus with $N_a \to \infty$ independent admissibility constraints, the set of globally admissible composite configurations localizes to a measure-

> zero subset of the composite phase space $\mathscr{M}\_\Phi$^(total). Moreover, this measure-zero subset consists of configurations in which the system topological sector and the apparatus pointer configuration are in definite correspondence: each admissible configuration has a definite (system winding sector, apparatus pointer) pair.

***Proof.*** *The measure-zero statement follows immediately from equation (7.3) as $N_a \to \infty$. The definite correspondence claim follows from the structure of the measurement interaction: by Definition 6.1, the coupling between $\Phi_s$ and $\Phi_a$ is such that the apparatus pointer configuration is admissibility-correlated with the system topological sector. For $N_a$ large, the admissibility constraints on the apparatus pointer configurations in sectors other than the one correlated with the surviving system sector are violated with probability $\exp(-\beta N_a)$ for some $\beta > 0$ derived from the measurement Hamiltonian (emergent description, Paper 2 [42]) and the apparatus construction. In the limit $N_a \to \infty$, these are exactly zero measure.* ∎

Lemma 6.1 is the mathematical heart of the theory of measurement. It establishes that macroscopic apparatus, by virtue of their enormous constraint density, effectively enforce a definite correspondence between system topological sector and pointer configuration — without any collapse, without any discontinuous jump, and without any observer. The outcome is definite not because a collapse postulate makes it so, but because the overwhelming majority of composite configurations are globally inadmissible when the apparatus is macroscopic. What survives the admissibility filter is a unique, definite outcome.

An important subtlety arises regarding the independence assumption for the apparatus constraints. In a real physical apparatus, the constraints are not exactly independent; they are correlated by the apparatus's internal dynamics. However, for a macroscopic apparatus in a sufficiently complex internal state (i.e., one that is thermalized or near-thermalized), the effective independence of constraints holds to exponentially good approximation by the central limit theorem applied to the constraint structure. Corrections to the independence assumption are of order

$N_a^{-1/2}$, which become negligible as $N_a \to \infty$, and they do not affect the measure-zero localization result of Lemma 6.1.

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## 8. Admissibility Pruning — The Mechanism of Outcome Selection

Having established in Lemma 6.1 that macroscopic apparatus localizes admissible composite configurations to a measure-zero subset, we now describe how this localization occurs dynamically — that is, how the admissibility pruning process unfolds as the system-apparatus coupling develops. The central claim is that this process is continuous, deterministic, and does not involve any discontinuous collapse.

Consider a system $\Phi_s$ in an initial configuration corresponding to a superposition of two topological winding sectors — call them $n = +1$ and $n = -1$ (a two-outcome measurement for definiteness; the generalization to K outcomes is straightforward). The apparatus $\Phi_a$ is initialized in a "ready" configuration with pointer in the neutral position. At time $\lambda = 0$ (where $\lambda$ is the parameter along the admissibility-constrained phase trajectory), the coupling between $\Phi_s$ and $\Phi_a$ is switched on. This coupling modifies the admissibility functional from $\mathscr{I}[\Phi_s] + \mathscr{I}[\Phi_a] + \mathscr{I}[\Phi_e]$ to $\mathscr{I}[\Phi_{total}]$ as given by equation (6.5), with the interaction terms $\mathscr{I}_{int}$ growing from zero as the coupling increases.

Let $\Omega_+(\lambda)$ and $\Omega_-(\lambda)$ denote the sets of composite configurations consistent with system sector $n = +1$ and $n = -1$ respectively, and satisfying $\mathscr{I}[\Phi_{total}(\lambda)] \geq 0$ at coupling strength $\lambda$. At $\lambda = 0$, both $\Omega_+(0)$ and $\Omega_-(0)$ are non-empty and have comparable admissibility measure. As $\lambda$ increases, the apparatus constraints become active: configurations in which the apparatus pointer is in the "+1 position" but the system is in sector $n = -1$ develop negative $\mathscr{I}_{int}$ contributions, pushing their $\mathscr{I}[\Phi_{total}]$ below zero. Similarly for the opposite mismatch. The dynamical equation governing this pruning is:

$$\frac{d}{d\lambda} \left[\ln \mu_{\mathscr{I}}(\Omega_j(\lambda))\right] = -\rho_c[\Phi_a] \cdot \kappa_j(\lambda) \quad (j = +, -) \tag{8.1}$$

where $\kappa_j(\lambda) \geq 0$ is a coupling-dependent pruning rate for sector j. For a measurement interaction, the pointer-sector correlation ensures that $\kappa_+(\lambda) >> 0$ for configurations inconsistent with the realized sector, and $\kappa_+(\lambda) = 0$ for the consistent sector. By Lemma 6.1, as $\lambda \to \infty$ (full coupling), one sector's measure vanishes exponentially while the other's remains stable (or approaches its admissibility measure maximum).

The key observation is that equation (8.1) is a continuous differential equation, not a jump condition. The admissibility measure of the "wrong" sector configurations decreases continuously and exponentially with increasing coupling. The outcome is determined not by a discontinuous collapse but by the exponential divergence of admissibility measures between competing sectors as the coupling grows. The apparent suddenness of measurement — the fact that on laboratory timescales the outcome seems instantaneous — is explained by the exponential rate of divergence: for $N_a \sim 10^{23}$, the time required for the losing sector's measure to fall below any detection threshold is of order the decoherence timescale $\tau_{dec}$, which for laboratory systems is typically $10^{-30}$ seconds or less.

To make this concrete, we work through a schematic two-outcome spin measurement. System $\Phi_s$ is in a phase configuration corresponding to a superposition of spin-up ($n = +½$) and spin-down ($n = -½$) winding sectors. The apparatus is a Stern-Gerlach-type device; its phase configuration $\Phi_a$ includes the magnetic field gradient (emergent from phase curvature, Paper 4 [44]) and a position-sensitive detector. The coupling produces admissibility correlations: configurations in which the spin-up component is paired with the "deflected-up" pointer state have $\mathscr{I}_{int} > 0$ (admissibility-favored) while configurations with spin-up paired with deflected-down have $\mathscr{I}_{int} < -\mathscr{I}_{threshold} < 0$ (admissibility-forbidden). As the coupling proceeds, the admissibility measure of the mismatched configurations is suppressed by $\exp(-\alpha' N_a)$, leaving — to overwhelming accuracy — only the matched pair: a definite spin outcome correlated with a definite pointer deflection.

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## 9. Outcome Selection Without Collapse

We now state and prove the central positive result of this paper.

**Proposition 6.1 (Outcome Selection Without Collapse).**

In a measurement interaction as defined by Definition 6.1, the outcome of measurement corresponds to the unique surviving admissible topological resolution of the composite $\Phi_{total} = \Phi_s \oplus \Phi_a \oplus \Phi_e$. This resolution is determined by global admissibility $\mathscr{I}[\Phi_{total}] \geq 0$ alone. No collapse of $\Phi_s$ occurs: the system phase configuration reorganizes as part of the composite, and the surviving configuration has a definite system topological sector not because of any primitive jump, but because all configurations with indefinite or mismatched sectors are globally inadmissible for macroscopic apparatus (Lemma 6.1). The result is a definite outcome that is unique, not one of many branches, and the mechanism is continuous admissibility constraint satisfaction.

***Proof.*** *Let the pre-measurement system phase configuration $\Phi_s^{(0)}$ correspond to a superposition of K topological winding sectors $n_1, \ldots, n_K$. By Section 8, as the system–apparatus coupling develops, admissibility pruning exponentially suppresses composite configurations corresponding to all but one sector–pointer pair. Specifically, for each pair $(n_j, \pi_j)$ where $\pi_j$ is the pointer state correlated with sector $n_j$, the admissibility measure of composite configurations outside the surviving pair satisfies $\mu_{\mathscr{I}}(\Omega_j(\lambda_f)) \leq \exp(-\alpha' N_a) \to 0$ as $N_a \to \infty$ (Lemma 6.1). In the limit $N_a \to \infty$ (completed measurement), exactly one pair $(n_{k^*}, \pi_{k^*})$ has non-zero admissibility measure. The composite $\Phi_{total}$ therefore has a definite system sector $n_{k^*}$ and a definite pointer configuration $\pi_{k^*}$. No collapse postulate was used; no jump occurred; only admissibility pruning operated.* ∎

Proposition 6.1 requires careful comparison with the Everett interpretation, since both claim to preserve unitarity (or its Phase Theory analog, admissibility-governed determinism) while yielding definite outcomes. The difference is fundamental. In the Everett interpretation, all K outcomes exist simultaneously in K branches of the universal wavefunction; the definite outcome experienced by an observer is a branch-relative fact, and all branches are equally real. In Phase Theory, there is only *one* globally admissible resolution — one outcome. The other $K-1$ outcomes are not realized in parallel branches; they are globally inadmissible configurations that do not exist. The topological stability argument (Section 10) shows why: topological configurations are binary — either topologically stable (winding number well-defined, admissible) or unstable (winding number ill-defined, inadmissible). There is no sense in which an inadmissible configuration "occurs in another branch." It simply does not occur.

Proposition 6.1 also requires comparison with decoherence-based accounts. Decoherence (Section 12) suppresses off-diagonal terms in the density matrix but, within standard unitary quantum mechanics, does not select one outcome over another; the post-decoherence state is an improper mixture that must be supplemented by a collapse postulate to yield a definite outcome. Proposition 6.1 goes further: within Phase Theory, admissibility pruning selects one outcome and eliminates all others from the space of physically realized configurations. No additional postulate is needed because admissibility is already a selection principle, not merely a suppression of interference.

Finally, we note that Proposition 6.1 implies that the question "which outcome occurs?" is not answered by Phase Theory through a stochastic process. The outcome is determined by the full initial condition of the composite — the complete global phase configuration $\Phi\_total(0)$ at the start of the measurement interaction. Since this global configuration is, in general, not accessible to any subcomponent of the composite, the outcome appears random to observers within the system. Its apparent probability is given by the admissibility measure of the initial configuration in the relevant sector, which reproduces the Born rule (Section 13). But this randomness is epistemic, not ontic: it reflects incomplete knowledge of the global phase configuration, not fundamental indeterminacy in the physical world.

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## 10. The Topology of Discrete Outcomes

The discreteness of measurement outcomes is one of the most striking features of quantum mechanics and a central explanandum of any foundational theory. In Phase

Theory, discreteness requires no special postulate: it is a direct consequence of the topological structure of phase configuration space established in Paper 3 [43].

Phase configurations are organized into topological sectors $\mathscr{M}_\Phi^{(n)} \subset \mathscr{M}_\Phi$, each characterized by a winding number vector $\mathbf{n} \in \mathbb{Z}^k$. Since winding numbers are integers by definition — they count how many times the phase winds around a topological cycle in $\mathscr{M}_\Phi$ — the topological sectors are discrete. There is no configuration with winding number $n = 1.5$; the space of configurations splits cleanly into sectors labeled by integers. Within a topological sector, the admissibility functional varies smoothly; between sectors, it must pass through a region of negative $\mathscr{I}$ — a "topological barrier" — which is globally forbidden.

This topological discreteness has several important consequences for measurement theory. First, the outcomes of a measurement are in bijection with the topological sectors accessible to the system phase configuration — that is, with the set of integers $\{n_1, \ldots, n_K\}$ corresponding to the sectors that appear in the initial system configuration. Second, the outcome cannot be continuously interpolated: there is no physical process within Phase Theory that continuously moves the system from winding sector $n = +1$ to winding sector $n = -1$ while remaining admissible. All paths between sectors pass through configurations with $\mathscr{I} < 0$, which are inadmissible. Third — and most importantly for the measurement problem — the discreteness of outcomes is basis-independent. The topological sectors are defined geometrically on $\mathscr{M}_\Phi$ and do not depend on any choice of basis in a Hilbert space. This resolves the *preferred basis problem* of the Everett interpretation.

The preferred basis problem in Everett is the question of what determines the preferred decomposition of the universal wavefunction into branches — what selects, for example, position eigenstates over momentum eigenstates as the "classical" branches. Many-worlds interpretations require an additional criterion (einselection, decoherence) to select the preferred basis. In Phase Theory, no such additional criterion is needed: the discrete topological sectors are defined independently of any basis choice, and the admissibility pruning process of Section 8 selects among these pre-defined sectors. The outcome basis is determined by the topology of $\mathscr{M}_\Phi$ relative to the apparatus coupling — it is a physical feature of the measurement setup, not a free choice.

Formally, the discreteness of outcomes follows from the following topological result. Let the system phase configuration space admit a decomposition $\mathscr{M}_\Phi^{(s)} = \coprod_n \mathscr{M}_\Phi^{(s,n)}$ into topological sectors (a coproduct of connected components). The admissibility constraint $\mathscr{I}[\Phi_{total}] \geq 0$ is compatible with this decomposition in the sense that for any globally admissible composite configuration $\Phi_{total} \in \mathscr{A}_{total}$, the system component $\Phi_s$ lies within a definite topological sector $\mathscr{M}_\Phi^{(s,n_k)}$ for some $n_k \in \mathbb{Z}^k$ (this follows from Lemma 6.1 and the topological structure of Paper 3 [43]). Hence measurement outcomes, as admissible resolutions of the composite, are necessarily discrete.

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## 11. Apparent Irreversibility and Admissibility Asymmetry

Measurement in standard quantum mechanics is irreversible: once a quantum system has been measured and has collapsed to an eigenstate, restoring the original superposition requires erasing all records of the measurement result throughout the environment — a practically impossible task. This irreversibility seems to conflict with the time-reversal symmetry of the Schrödinger equation, and its explanation has been a source of ongoing debate. Phase Theory provides a clean and quantitative resolution.

**Proposition 6.2 (Irreversibility as Admissibility Asymmetry).**

The apparent irreversibility of measurement is not a fundamental asymmetry in Phase Theory but is a consequence of the combinatorial admissibility suppression of the time-reversed composite configuration. Formally, let $\Phi_{total}^{(f)}$ be the post-measurement composite phase configuration. Its time-reversed configuration $\Phi_{total}^{(R)} = \Phi_{total}^{(f)}|_{\{\lambda \to -\lambda\}}$ (reversal of the phase trajectory parameter) satisfies $\mathscr{I}[\Phi_{total}^{(R)}] < 0$ with probability approaching 1 as $N_a \to \infty$.

***Proof sketch.*** *The time-reversed configuration $\Phi_{total}^{(R)}$ requires the apparatus pointer to move from its post-measurement definite position back to the neutral position, coordinated with the reversal of $O(N_a)$ internal apparatus degrees of* *freedom and the reversion of $O(N_e)$ environmental degrees of freedom correlated with the measurement. The admissibility constraints on such a coordinated reversal configuration are the same constraints as on the forward configuration, but the forward evolution was driven by these constraints (the system-apparatus coupling). The reverse evolution requires all $N_a$ apparatus degrees of freedom to coordinately evolve against the direction favored by the admissibility constraints. The probability of finding a globally admissible configuration with this coordinate reversal property is bounded above by $\exp(-\beta' N_a)$ for some $\beta' > 0$, by the same argument as equation (7.3) applied to the reversal sector. As $N_a \to \infty$, this probability vanishes.* ∎

Proposition 6.2 establishes that measurement irreversibility is not a primitive fact about time but a derived consequence of the combinatorial overwhelmingness of the apparatus constraint system. This is closely analogous to, but more fundamental than, the thermodynamic explanation of the second law: the second law arises in standard statistical mechanics from the enormous number of microstates corresponding to high-entropy macrostates. In Phase Theory, the analogous statement is that the enormous number of admissibility constraints imposed by a macroscopic apparatus make the reverse trajectory overwhelmingly inadmissible.

The connection to thermodynamic irreversibility is direct. In Phase Theory, entropy is an admissibility measure concept: the entropy of a macroscopic phase configuration is (up to a constant) the logarithm of the admissibility measure of the macroscopic configurations consistent with the observed macroscopic state. The second law of thermodynamics — that entropy tends to increase — corresponds in Phase Theory to the statement that admissibility-governed trajectories tend toward configurations with higher admissibility measure (higher entropy), because lower-measure configurations are progressively pruned by the global constraint. Measurement irreversibility is the special case of this general tendency in which the "entropy" in question is the admissibility measure of the measurement pointer sector — and the post-measurement pointer state corresponds to a very high-measure macroscopic configuration.

A subtlety worth noting: Proposition 6.2 shows that reversal has probability $\exp(-\beta' N_a)$, not exactly zero for finite $N_a$. This means that for very small $N_a$ — a few-particle "apparatus" — measurement can be approximately reversible. This is consistent with known quantum physics: quantum erasure (Section 19) is possible for small systems. The transition from reversible to irreversible measurement occurs at around the macroscopicity threshold $\rho_*$ of Definition 6.1, and Phase Theory predicts that this threshold scales as $N_a \sim (\beta')^{-1} \ln(\varepsilon^{-1})$ for a desired reversal suppression $\varepsilon$, a testable prediction.

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## 12. Decoherence Reinterpreted as Admissibility Collapse of Cross-Terms

The decoherence program, initiated by Zeh [9] and developed by Zurek [8, 46], Joos et al. [10], and Schlosshauer [39, 40], establishes that the interaction of a quantum system with a macroscopic environment suppresses the off-diagonal terms of the system's reduced density matrix on a timescale $\tau_{dec}$ that, for macroscopic objects, is extraordinarily short. This accounts for the empirical absence of observable quantum superpositions of macroscopically distinct states. Phase Theory subsumes and extends the decoherence program: the suppression of off-diagonal terms is a special case of admissibility pruning of cross-term configurations.

In standard quantum mechanics, the off-diagonal term $\rho_{ij} = \langle i|\rho_s|j\rangle$ of the reduced density matrix represents quantum coherence between outcome states $|i\rangle$ and $|j\rangle$. Decoherence suppresses $\rho_{ij} \to 0$ due to entanglement between the system and environment. In Phase Theory language, the off-diagonal term corresponds to a composite phase configuration $\Phi_{total}$ that is a topological "hybrid" between sector $n_i$ and sector $n_j$ — a configuration that is neither fully in one winding sector nor the other but involves a phase excitation interpolating between them. Such interpolating configurations generically have a phase mismatch at the system–environment boundary: the system sector (partially $n_i$, partially $n_j$) does not match any definite environmental configuration in a globally admissible way, because the environment's pointer-like degrees of freedom (the "environmental registers" of Zurek [8]) are already correlated with definite sector assignments through their internal dynamics.

Formally, a cross-term configuration $\Phi_{cross}^{ij}$ (one contributing to the off-diagonal $\rho_{ij}$) satisfies:

$$(12.1)\quad \mathscr{I}_{int}[\Phi_s^{cross}, \Phi_e] < -\delta_{ij} < 0$$

where $\delta_{ij} > 0$ depends on the system–environment coupling strength and the topological distance between sectors $n_i$ and $n_j$. Equation (12.1) ensures that $\mathscr{I}[\Phi_{total}^{cross}] = \mathscr{I}[\Phi_s^{cross}] + \mathscr{I}[\Phi_e] + \mathscr{I}_{int}[\ldots] < 0$ for sufficiently large $N_e$, making these configurations globally inadmissible. The timescale on which cross-term configurations become inadmissible — the admissibility pruning timescale $\tau_{prune}$ — corresponds exactly to the decoherence timescale $\tau_{dec}$:

$$(12.2)\quad \tau_{prune} \leq \tau_{dec}$$

***Derivation sketch of (12.2).*** *The decoherence timescale $\tau_{dec}$ is determined by the rate at which environmental entanglement accumulates (Zurek [8], Schlosshauer [39]): $\tau_{dec} \sim \tau_{relax} / (\Delta x/x_{th})^2$ for spatial superpositions, where $\tau_{relax}$ is the relaxation time and $x_{th}$ is the thermal de Broglie wavelength. In Phase Theory, the admissibility pruning rate for cross-term configurations follows from equation (8.1): $d/d\lambda[\ln \mu_{\mathscr{I}}(\Omega_{cross}(\lambda))] = -\rho_c[\Phi_e] \cdot \kappa_{cross}(\lambda)$. Since $\rho_c[\Phi_e]$ is proportional to the environmental coupling rate, and $\kappa_{cross}$ is proportional to the phase mismatch $\delta_{ij}$, the pruning rate equals the decoherence rate and $\tau_{prune} = \tau_{dec}$. The inequality ≤ arises because admissibility pruning is the more general process; decoherence is its coherent-limit description.* ∎

The conceptual advance of Phase Theory over the standard decoherence program is significant. Standard decoherence, applied within unitary quantum mechanics, produces an improper mixed state — a density matrix diagonal in the pointer basis, but representing a superposition of all outcomes, not a definite one. This leaves the measurement problem unresolved. Phase Theory's admissibility pruning goes further: it eliminates the cross-term configurations from physical reality (not merely from the reduced density matrix), and selects a unique outcome via Proposition 6.1. Decoherence is thus a necessary but not sufficient component of the measurement mechanism; admissibility selection is the additional element that completes the story without any new postulate.

## 13. The Born Rule from Admissibility Measure

The Born rule — that the probability of obtaining outcome $n_k$ upon measurement is $P(n_k) = |\psi_k|^2$ where $\psi_k = \langle n_k|\psi\rangle$ is the complex amplitude — is standardly postulated as an irreducible axiom of quantum mechanics. Gleason's theorem [24] shows that the Born rule is the unique probability assignment consistent with the Hilbert space structure of quantum mechanics, but this is a uniqueness proof, not a derivation from more primitive principles. In Phase Theory, the Born rule is derived from the admissibility measure $\mu_{\mathscr{I}}$, established in Paper 5 [45] as the fundamental probability concept.

**Theorem 6.2 (Born Rule Derivation).**

Let $\Phi_s$ be a system phase configuration in the coherent limit — that is, a configuration admitting a description in terms of complex phase amplitudes $\psi_k$ associated with topological winding sectors $n_k$, $k = 1, \ldots, K$. Then the admissibility measure $\mu_{\mathscr{I}}$ of the phase-space region corresponding to configurations resolving to winding sector $n_k$ satisfies:

$$\mu_{\mathscr{I}}(\text{sector } n_k \mid \Phi_s) = |\psi_k|^2 \tag{13.1}$$

where $|\psi_k|^2$ is computed from the standard quantum mechanical inner product in the Hilbert space limit of Phase Theory.

***Proof.*** *The proof proceeds in three steps, drawing on Paper 5 [45].* ***Step 1:*** *Paper 5, Theorem 5.4, establishes that in the coherent limit, the admissibility measure $\mu_{\mathscr{I}}$ on $\mathscr{M}_\Phi$ reduces to a measure on the space of complex amplitudes $\mathscr{P} = \{(\psi_1, \ldots, \psi_K) \in \mathbb{C}^K : \Sigma|\psi_k|^2 = 1\}$ (the standard quantum state space, viewed as a real manifold $CP^{K-1}$). Step 2: Paper 5, Proposition 5.7, establishes that $\mu_{\mathscr{I}}$ on $\mathscr{P}$ is the Fubini-Study measure on $CP^{K-1}$, which is the unique $U(K)$-invariant measure on the complex projective space. Step 3: The admissibility measure of the sector $n_k$ relative to the state $(\psi_1, \ldots, \psi_K)$ is the $\mu_{\mathscr{I}}$-measure of configurations in which $n_k$ is selected* *by admissibility pruning. By Proposition 6.1 and Lemma 6.1, in the $N_a \to \infty$ limit, the fraction of initial phase configurations leading to outcome $n_k$ is the Fubini-Study measure of the region $\{(\varphi_1,\ldots,\varphi_K) : |\varphi_k|^2 \geq |\varphi_j|^2 \text{ for all } j \neq k\}$ averaged over the Fubini-Study distribution centered on $(\psi_1,\ldots,\psi_K)$. Standard calculation (Paper 5, Theorem 5.5) yields that this equals $|\psi_k|^2$, establishing equation (13.1).* ∎

Theorem 6.2 establishes that the Born rule is not a postulate but a theorem within Phase Theory, valid in the coherent limit. Let us be precise about what "coherent limit" means. The coherent limit is the regime in which the system phase configuration is dominated by a single coherently propagating phase mode — the regime in which standard quantum mechanical wavefunctions are a good approximation to the full phase configuration. In this regime, the full admissibility measure reduces to the Fubini-Study measure on complex amplitudes, and equation (13.1) holds exactly. Outside the coherent limit — in regimes of strong topological non-linearity, or in the very early universe where phase configurations are highly non-coherent — Phase Theory predicts deviations from the Born rule. These deviations are among the novel predictions discussed in Section 25.

An important connection: Gleason's theorem [24] states that the only non-contextual probability assignment on the lattice of closed subspaces of a Hilbert space of dimension ≥ 3 is the Born measure $P(\hat{P}) = \mathrm{Tr}(\rho\, \hat{P})$. Corollary 6.2 below shows that Gleason's theorem is recovered from Phase Theory as a special case of admissibility measure uniqueness.

**Corollary 6.2 (Gleason's Theorem as Admissibility Uniqueness).**

Gleason's theorem is a consequence of the uniqueness of the admissibility measure $\mu_{\mathscr{I}}$ on the coherent sector of phase configuration space. Specifically, the U(K)-invariance of the Fubini-Study measure (established in Paper 5 [45]) implies that it is the unique probability measure on $CP^{K-1}$ invariant under all admissible phase reparametrizations, which in the coherent limit correspond to unitary transformations. The non-contextual probability assignment of Gleason follows immediately.

The derivation of the Born rule in Theorem 6.2 addresses one of the deepest difficulties in the Everett interpretation — the probability problem. In Everett, if all outcomes occur in parallel branches, it is unclear why the probability of "ending up" in branch k should be $|\psi_k|^2$ rather than 1/K (equal branch weighting) or any other measure. Various approaches (decision-theoretic derivations by Deutsch, Wallace; branch counting arguments) have been proposed but remain controversial. In Phase Theory, this difficulty does not arise: there is only one outcome (Proposition 6.1), and its probability is the admissibility measure of the initial configuration in the relevant sector, which is demonstrably $|\psi_k|^2$ by Theorem 6.2.

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## 14. No Observer Privilege — Observers as Phase Structures

One of the most philosophically fraught aspects of quantum mechanics is the apparent role of the observer. In some formulations — notably the original Copenhagen interpretation and von Neumann's chain — the observer or observer's consciousness plays an indispensable role in collapsing the wavefunction and producing a definite outcome. Phase Theory eliminates observer privilege entirely.

**Proposition 6.3 (No Observer Privilege).**

Let $\Phi_{obs}$ denote the phase configuration of an observer — a macroscopic phase structure capable of registering and storing measurement outcomes. The inclusion of $\Phi_{obs}$ in the composite measurement interaction — replacing $\Phi_e$ by $\Phi_e \oplus \Phi_{obs}$ — changes the admissibility constraint to $\mathscr{I}[\Phi_s \oplus \Phi_a \oplus \Phi_e \oplus \Phi_{obs}] \geq 0$, adding further constraints but changing nothing in kind. In particular: (a) the observer's presence does not alter the set of possible outcomes; (b) the observer's presence does not alter the admissibility measure of outcomes; (c) the observer's consciousness introduces no new physical mechanism; (d) measurement occurs regardless of whether a conscious observer is present.

***Proof.*** *By Definition 6.1, measurement is defined by the macroscopic character of the apparatus ($\rho_c[\Phi_a] \geq \rho_*$) and the system–apparatus admissibility correlation, not by the presence of an observer. By Lemma 6.1, outcome selection occurs via admissibility pruning by the apparatus constraints, which is complete by the time the measurement interaction ends — before the observer registers the result. Adding $\Phi_{obs}$ to the composite introduces additional admissibility constraints ($\rho_c[\Phi_{obs}] > 0$), but these constraints act on the observer's memory state, not on the system-apparatus outcome. The system-apparatus admissibility correlation is determined by the apparatus constraint density $\rho_c[\Phi_a]$ alone. Statements (a)–(d) follow immediately.* ∎

Proposition 6.3 enables a Phase Theory analysis of the Wigner's Friend scenario, which we defer to Section 16 for full treatment. Here we note one consequence: the von Neumann chain dissolves. Von Neumann's chain traced the need for a conscious observer through an arbitrarily long sequence of measuring devices, each measuring the next, and concluded that the chain terminates only at consciousness. In Phase Theory, the chain terminates not at consciousness but at the first macroscopic apparatus with $\rho_c[\Phi_a] \geq \rho_*$. Any macroscopic apparatus — whether or not connected to a conscious observer — performs outcome selection via admissibility pruning. A photographic plate, a Geiger counter, a cloud chamber track — all of these have sufficient constraint density to effect complete admissibility pruning, producing definite outcomes without any observer.

The role of consciousness in physics has been the subject of extended debate, from von Neumann's chain through Wigner's explicit invocation of mind-body interaction to Stapp's quantum mind theory. Phase Theory's position is unambiguous: consciousness is a macroscopic phase structure (an emergent property of sufficiently complex, highly organized phase configurations with high internal admissibility margin), and it plays no causal role in measurement beyond that of any other macroscopic phase structure. This is not a claim that consciousness is unimportant or uninteresting — it may well be the most complex emergent structure in Phase Theory — only that it is not ontologically privileged with respect to the admissibility constraint. The measurement problem, insofar as it required invoking consciousness, was pointing to an error in the identification of primitives, not to a genuine role for mind in physics.

## 15. Contextuality — Structural, Not Mysterious

Contextuality is the quantum mechanical phenomenon by which the outcome statistics of a measurement depend on what other measurements are performed simultaneously — on the "context" of the measurement. The Kochen-Specker theorem [14] establishes that it is impossible to assign definite pre-measurement values to all quantum observables simultaneously in a context-independent way; Bell inequalities [12] establish that local hidden variable theories cannot reproduce quantum correlations. Contextuality has been regarded as one of the most mysterious features of quantum mechanics. Phase Theory explains it as a structural consequence of global admissibility non-separability.

**Corollary 6.1 (Contextuality from Non-Separability).**

Kochen-Specker contextuality is a corollary of the non-separability of the admissibility functional $\mathscr{I}[\Phi_{total}]$ established in Proposition 6.0 (equation 6.5). Specifically: different apparatus configurations $\Phi_a^{(1)}$, $\Phi_a^{(2)}$ impose different admissibility constraints on the composite, yielding different surviving configurations and hence different outcome statistics for the system $\Phi_s$, even when the system itself is unchanged. This context-dependence of statistics is an inevitable consequence of global admissibility and does not require hidden variables or non-locality in any mysterious sense.

***Proof.*** *By equation (6.5), $\mathscr{I}[\Phi_s \oplus \Phi_a^{(1)} \oplus \Phi_e]$ and $\mathscr{I}[\Phi_s \oplus \Phi_a^{(2)} \oplus \Phi_e]$ differ by the interaction terms $\mathscr{I}_{int}[\Phi_s, \Phi_a^{(1)}] \neq \mathscr{I}_{int}[\Phi_s, \Phi_a^{(2)}]$ for $\Phi_a^{(1)} \neq \Phi_a^{(2)}$. Hence the sets of globally admissible configurations $\mathscr{A}_{total}^{(1)}$ and $\mathscr{A}_{total}^{(2)}$ differ, and by Proposition 6.1, the outcomes selected by admissibility pruning differ. The admissibility measure of sector $n_k$ under apparatus (1) vs (2) need not be equal. This is Kochen-Specker contextuality: outcome statistics depend on context (apparatus). The derivation uses only non-separability and requires no additional postulate.* ∎

The explanatory power of Corollary 6.1 is significant. Kochen-Specker contextuality has often been presented as showing that quantum mechanics cannot be completed by hidden variables: no pre-existing values can be assigned to observables. From the Phase Theory perspective, this result is unsurprising. There are no "hidden values" to assign because the outcomes are not pre-determined by the system alone; they are determined by the global admissibility of the full composite $\Phi_{total}$, which includes the apparatus. Changing the apparatus changes the admissibility constraints and hence can change the outcome statistics. This is structural, not mysterious: it follows necessarily from the global character of the admissibility constraint.

Phase Theory also explains why contextuality does not enable superluminal signaling. Different apparatus contexts change the local outcome statistics of system measurements, but they cannot change the reduced admissibility measure of a spatially remote subsystem, because the interaction terms $\mathscr{I}_{int}$ are bounded by the admissibility functional and its emergent causal structure (Paper 2 [42]). The no-signaling constraint is built into the structure of global admissibility; it is not imposed separately.

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## 16. Wigner's Friend and Nested Measurement

The Wigner's Friend thought experiment [17] presents a putative paradox about nested measurement. Friend F measures system S, obtaining a definite outcome. From F's perspective, S has collapsed to a definite state. But Wigner W, outside the laboratory, applies the unitary Schrödinger evolution to the composite (S + F + lab) and finds it in a superposition of the two measurement outcomes. Who is correct? When did collapse occur? Phase Theory dissolves this paradox completely.

Let us formalize the scenario in Phase Theory language. The initial state of the system is $\Phi_s^{(0)}$, corresponding to a superposition of winding sectors $n = +1$ and $n = -1$. Friend F is represented by the phase configuration $\Phi_F$ of a macroscopic observer. Wigner is represented by $\Phi_W$. The laboratory environment is $\Phi_{lab}$. The sequence of events is as follows:

First, F performs a measurement of S within the laboratory. By Proposition 6.1 and Lemma 6.1, since $\Phi_F$ is macroscopic ($\rho_c[\Phi_F] \geq \rho_*$), admissibility pruning selects a definite outcome — say $n = +1$. The post-measurement composite configuration $\Phi_{(S+F+lab)}^{(1)}$ has S in sector $n = +1$ and F's memory in the "observed +1" state. This is globally admissible: $\mathcal{I}[\Phi_{(S+F+lab)}^{(1)}] \geq 0$. The "$n = -1$ with F observing +1" configuration has been eliminated by admissibility pruning of F's macroscopic constraint set.

Second, Wigner W measures the composite (S + F + lab). In standard quantum mechanics, if W applies a unitary superposition measurement, he finds the composite in a superposition, apparently contradicting F's definite outcome. In Phase Theory, W's measurement is a further phase coupling: $\Phi_{total}^{(2)} = \Phi_{(S+F+lab)}^{(1)} \oplus \Phi_W$. The key point is that $\Phi_{(S+F+lab)}^{(1)}$ is already in a definite topological sector — $n = +1$ with F's memory in the "+1 observed" configuration. W's admissibility pruning therefore acts on a composite that is already in a definite sector. W's measurement outcome is the definite sector of $\Phi_{(S+F+lab)}^{(1)}$, not a superposition.

The apparent paradox in standard quantum mechanics arises because Wigner treats the composite (S + F + lab) as being in a quantum superposition after F's measurement — applying the unitary Schrödinger equation globally and finding that no collapse has occurred from Wigner's external perspective. In Phase Theory, this is not paradoxical: F's measurement *did* produce a definite outcome (via admissibility pruning at F's scale), and this outcome is encoded in the global phase configuration. Wigner's "superposition" description of the post-F-measurement composite is not an accurate description of the global phase configuration; it is an artifact of ignoring the admissibility constraints that F's macroscopic apparatus has already satisfied.

> **Proposition (Wigner's Friend Consistency).**
>
> Phase Theory predicts the same experimental outcomes as standard quantum mechanics for all Wigner's Friend protocols proposed to date — including those of Brukner [see 33] and Frauchiger-Renner — while providing a consistent ontology: F's definite outcome and W's measurement are not contradictory descriptions of

> the same reality, but descriptions at different scales of constraint density. No paradox arises because global admissibility is unique: there is only one evaluation, $\mathscr{I}[\Phi_{total}^{(2)}] \geq 0$, and it selects a unique definite outcome for W that is consistent with F's prior definite outcome.

The key ontological resolution of the Wigner's Friend paradox in Phase Theory is this: there is no "Wigner's perspective" and "Friend's perspective" that can genuinely disagree about the physical reality of the system's state. There is only one global phase configuration, one global admissibility evaluation, and one definite outcome. "Perspectives" in Phase Theory are not windows onto different realities but are coarse-grainings of the same global admissible configuration at different scales of constraint density. The Friend's coarse-graining (at the scale of the Friend-system-lab composite) and Wigner's coarse-graining (at the scale of the full composite including the lab) both represent the same globally admissible configuration; they are consistent by construction.

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## 17. Delayed-Choice Experiments and Temporal Admissibility

Wheeler's delayed-choice experiment [16] is among the most striking demonstrations of quantum non-classicality. In the standard version, a photon passes through a double-slit apparatus; the experimenter then freely chooses, *after* the photon has passed through the slits, whether to observe which-slit information or whether to observe interference. If which-slit is observed, no interference pattern appears; if interference is observed, an interference pattern appears — as if the photon "knew in advance" what measurement would be performed. This seems to imply that the future choice of measurement retroactively determines the photon's past behavior. Phase Theory dissolves this apparent retrocausality.

In Phase Theory, the photon is not a primitive object traveling through space (Paper 3 [43]). It is a propagating phase excitation — a topological mode of the phase field $\Phi$ traversing the configuration space $\mathscr{M}_\Phi$. The double-slit apparatus is a phase configuration $\Phi_{slits}$ that introduces admissibility constraints on the excitation's topological routing. The detector configuration — which-slit detector or interference screen — is an apparatus phase configuration $\Phi\_det$. The composite is $\Phi\_total = \Phi\_excitation \oplus \Phi\_slits \oplus \Phi\_det \oplus \Phi\_environment$.

The admissibility constraint $\mathscr{I}[\Phi\_total] \geq 0$ is global in the sense that it encompasses the full composite, including both the "past" topology of the slit traversal and the "future" topology of the detection. Since admissibility in Phase Theory is not defined relative to any background time (time is an emergent coordinate, Paper 2 [42]), there is no sense in which the slit-traversal event is "earlier" than the detector configuration in any fundamental way. Admissibility is evaluated on the full composite simultaneously, and the surviving configurations are those consistent with the full apparatus arrangement — including the detector choice.

The concept of **Temporal Admissibility Consistency** formalizes this. Let $\Phi\_total = \Phi\_excitation \oplus \Phi\_slits \oplus \Phi\_det^{(choice)} \oplus \Phi\_environment$ be the composite for a given detector choice. The admissibility constraint $\mathscr{I}[\Phi\_total] \geq 0$ selects configurations consistent with this composite. If the choice is the interference screen ($\Phi\_det^{(int)}$), the admissible configurations include those with topological routing through both slits simultaneously (superposition of routing sectors), yielding interference. If the choice is the which-slit detector ($\Phi\_det^{(which)}$), the detector's admissibility constraints eliminate cross-routing configurations (cross-terms become inadmissible, as in Section 12), yielding no interference.

The apparent retrocausality arises from describing this in terms of a background time and a photon "deciding" its path. In Phase Theory, there is no such decision at any moment: the full composite configuration — including the detector — is admissibility-constrained globally, and the surviving configurations are those consistent with the full apparatus. The late-imposed detector constraint reshapes the globally admissible configurations of the composite without any causal influence propagating backward in time, because admissibility is not evaluated sequentially in time but simultaneously on the full phase space. This is consistent with the absence of signaling (the experimenter cannot use delayed choice to send information to the past), because the admissibility measure of outcomes accessible to any observer does not depend on the sequence in which constraints are "applied" — the global constraint is unique regardless of the order of description.

## 18. EPR Correlations, Bell Inequalities, and Non-Local Admissibility

The Einstein-Podolsky-Rosen argument [4] and Bell's subsequent inequalities [12] have established beyond any doubt that quantum mechanics cannot be explained by local hidden variable theories. Bell inequality violations have been confirmed experimentally to high precision [15], with recent loophole-free tests removing all significant experimental loopholes. These results require any foundational theory to either embrace non-locality (as Bohmian mechanics does), abandon realism (as Copenhagen and QBism do), or embrace many outcomes (as Everett does). Phase Theory provides a fourth resolution: the correlations arise from global admissibility constraints on the composite, which are non-local in emergent spacetime coordinates but local in phase configuration space.

> **Theorem 6.3 (Bell Correlations from Global Admissibility).**
>
> Phase Theory reproduces all Bell-inequality-violating correlations predicted by quantum mechanics — including the CHSH inequality bound violation — without hidden variables and without superluminal signaling, through global admissibility constraints on the composite phase configuration $\Phi_A \oplus \Phi_B \oplus \Phi_{environment}$, where A and B are spatially separated measurement events. Furthermore, the Tsirelson bound $\langle CHSH \rangle \leq 2\sqrt{2}$ emerges as the admissibility maximum of the CHSH correlator over all globally admissible composite configurations.

***Proof sketch.*** *Consider an entangled pair in the singlet phase configuration $\Phi_{sing} = \Phi_A \oplus \Phi_B$, corresponding in the coherent limit to the quantum state $(|+A-B\rangle - |-A+B\rangle)/\sqrt{2}$. This configuration has a definite global topological charge (total winding number zero for the singlet) enforced by $\mathscr{I}[\Phi_{sing}] \geq 0$. When A is measured by apparatus $\Phi_a^A$ and B is measured by apparatus $\Phi_a^B$, the composite is $\Phi_{total} = \Phi_{sing} \oplus \Phi_a^A \oplus \Phi_a^B \oplus \Phi_{environment}$. The global admissibility constraint enforces that the surviving configurations maintain the global topological charge: if A* *is measured in sector $n = +1$, then B's surviving configurations must be in $n = -1$ (for antiparallel settings). This global correlation is built into $\mathscr{I}[\Phi\_total] \geq 0$ and is independent of the spatial separation of A and B, because $\mathscr{I}$ is an integral over all of $\mathscr{M}\_\Phi^{(total)}$ and is not sensitive to emergent spatial coordinates. For non-antiparallel settings (angle $\theta$ between measurement directions), the admissibility measure of correlated outcomes gives the standard quantum prediction $\langle A \cdot B \rangle = -\cos \theta$. The CHSH correlator is $\langle CHSH \rangle = |\langle A\_1B\_1 \rangle + \langle A\_1B\_2 \rangle + \langle A\_2B\_1 \rangle - \langle A\_2B\_2 \rangle|$; maximizing over angles within the admissibility constraint gives $2\sqrt{2}$, the Tsirelson bound. No signaling occurs because the admissibility constraints do not couple the A-sector and B-sector marginals: changing $\Phi\_a^A$ does not change the admissibility measure of B-sector configurations marginally, only the joint correlations.* ∎

The key conceptual point of Theorem 6.3 is the distinction between non-locality in emergent spacetime and non-locality in phase configuration space. The global admissibility constraint $\mathscr{I}[\Phi\_total] \geq 0$ is "non-local" in the sense that it is an integral over the full composite space and thus correlates configurations associated with spatially separated events. But this non-locality is in phase configuration space — the fundamental ontological substrate — not in emergent spacetime. In emergent spacetime (Paper 2 [42]), the causal structure ensures no signaling: the admissibility constraints enforce correlations but not communication. This is exactly the kind of "non-local but non-signaling" structure that Bell's theorem shows is required for any realistic theory reproducing quantum correlations. Phase Theory provides it from first principles.

The Tsirelson bound $2\sqrt{2}$ has a natural interpretation in Phase Theory: it is the maximum CHSH correlator achievable by globally admissible configurations of the composite. Configurations with $\langle CHSH \rangle > 2\sqrt{2}$ — so-called "super-quantum" or Popescu-Rohrlich box correlations — correspond to composite configurations that are globally inadmissible (they would require topological configurations with no admissible phase winding structure). Phase Theory thus explains not just why quantum correlations violate the classical Bell bound of 2, but also why they do not violate the Tsirelson bound: the Tsirelson bound is an admissibility maximum, not merely a quantum mechanical coincidence.

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## 19. Quantum Erasure and Which-Path Information

The quantum eraser experiment [26] extends the delayed-choice setup by introducing an eraser: a device that can either record which-path information or erase it, with the detection conditioned on the eraser's action. When which-path information is present and not erased, no interference is observed. When the information is erased (before detection), interference is restored — even if the erasure decision is made after the photon has been detected. This seems to imply that future actions can retroactively affect past events. Phase Theory provides a clean, non-retrocausal explanation.

In Phase Theory language: which-path information corresponds to composite configurations in which the photon phase excitation has established admissibility correlations with a "path register" — an apparatus degree of freedom $\Phi_r$ that records which topological routing sector the excitation traversed. The presence of this correlation makes the cross-routing configurations (those contributing to interference) globally inadmissible: $\mathscr{I}_{int}[\Phi_{excitation}^{cross}, \Phi_r] < 0$ by equation (12.1). Hence interference is suppressed when which-path information is present in the composite.

When the eraser operates — that is, when the path register $\Phi_r$ is reset to a phase configuration that is uncorrelated with the routing sector — the interaction term $\mathscr{I}_{int}[\Phi_{excitation}, \Phi_r^{erased}]$ returns to zero (or close to zero, depending on the quality of the erasure). The cross-routing configurations, which were inadmissible when $\mathscr{I}_{int} < 0$, now satisfy $\mathscr{I}[\Phi_{total}^{erased}] \geq 0$ and are restored to the admissible set. The condition for erasure to restore interference is precisely:

$$\mathscr{I}[\Phi_{excitation}^{cross} \oplus \Phi_r^{erased} \oplus \Phi_{rest}] \geq 0 \quad (19.1)$$

Equation (19.1) is the Phase Theory erasure condition. It states that erasure restores interference when — and only when — the erased register configuration $\Phi_r^{erased}$ brings cross-term configurations back within global admissibility. This occurs when the erasure is complete: the register no longer retains any admissibility correlations with the routing sector.

The apparent retrocausality of quantum erasure — the fact that interference appears to be restored by an event (the erasure decision) made after detection — is dissolved in Phase Theory by the same argument as delayed choice (Section 17). The admissibility constraint $\mathscr{I}[\Phi\_total] \geq 0$ is evaluated globally on the full composite, which includes the eraser and its state. The global admissibility evaluation simultaneously determines whether cross-routing configurations are admissible or not; it is not a sequential temporal calculation. The eraser's state is part of the global phase configuration, and global admissibility selects configurations consistent with the full setup — erasure included — without any temporal ordering in the fundamental evaluation.

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## 20. Comparison with Objective Collapse Theories

Objective collapse theories — GRW [7], CSL (Continuous Spontaneous Localization) [38], and Penrose's objective reduction [8p, 37] — propose that the Schrödinger equation is modified by additional stochastic or gravitational terms that spontaneously localize wavefunctions, producing definite outcomes without observer intervention. These theories are empirically distinct from standard quantum mechanics and represent genuine scientific alternatives. We compare them with Phase Theory systematically.

| Property | GRW / CSL | Penrose OR | Phase Theory |
|---|---|---|---|
| **Ontological primitives** | Wavefunction + stochastic hits | Wavefunction + spacetime geometry | Phase configuration $\Phi$ only |
| **Collapse mechanism** | Spontaneous localization (GRW rate $\lambda$, width r_C) | Quantum gravity ($E_G > \hbar/\tau$) | Admissibility pruning (no new mechanism) |
| **Free parameters** | Two ($\lambda$, r_C in GRW); one additional in CSL | None (but requires quantum gravity theory) | None (all determined by $\mathscr{I}[\Phi]$) |
| **Unitarity** | Violated (by design) | Violated (by gravity) | Preserved (admissibility-governed) |
| **Lorentz invariance** | Problematic (GRW); addressed in relativistic CSL but not fully solved | Problematic | Emergent Lorentz symmetry from phase structure (Paper 2) |

| Property | GRW / CSL | Penrose OR | Phase Theory |
|---|---|---|---|
| **Observer role** | None (collapse is objective) | None | None (phase structure only) |
| **Deviations from standard QM** | Yes — energy non-conservation; heating; decoherence deviations | Yes — at Penrose scale ($E_G \sim \hbar/\tau$) | Yes — topology-dependent, very small (Section 25) |
| **Tested regimes** | Consistent with all current tests (LISA Pathfinder, DECIDE) | Not yet definitively tested | Consistent with all current tests |

Phase Theory's most significant advantage over GRW and CSL is the absence of free parameters. The GRW collapse rate $\lambda \sim 10^{-16}\ s^{-1}$ and localization width $r_C \sim 10^{-7}$ m must be chosen to make collapse too rare to affect individual microscopic systems while fast enough to produce effectively classical macroscopic behavior. These values are not derived from any deeper principle; they are fitted to the phenomenology. Phase Theory derives all measurement phenomenology from the admissibility functional $\mathscr{I}[\Phi]$, which itself is fixed by the axioms of Phase Theory. The equivalent of $\lambda$ and $r_C$ in Phase Theory — the constraint density threshold $\rho_*$ and the admissibility pruning rate $\alpha'$ — are determined in principle by the admissibility functional, not freely chosen.

Phase Theory could in principle be distinguished from CSL experimentally through the N-dependence of decoherence suppression. CSL predicts a specific mass-density-based collapse rate; Phase Theory predicts an admissibility-based pruning rate $\tau_{prune} \sim N^{-\gamma}$ (Section 25). In regimes of moderate N (100–10,000 particles), the two predictions diverge at a level potentially accessible to next-generation matter-wave interferometry experiments (macromolecule or nanoparticle interferometers). The DECIDE experiment [see 33, 38] is an early step in this direction; Phase Theory predicts no CSL-type signal in the mass range probed by DECIDE, consistent with current null results, but predicts specific signatures in the N-dependence of decoherence onset at higher mass.

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## 21. Comparison with No-Collapse Interpretations

We compare Phase Theory with the three principal no-collapse interpretations: Everett Many-Worlds [5, 49], Bohmian Mechanics [11], and QBism [13]. All three preserve the unitarity of quantum evolution but differ radically in their ontological commitments. Phase Theory also preserves admissibility-governed (effectively unitary) evolution while differing from all three in ontology and in some empirical predictions.

**Everett Many-Worlds.** Everett's relative-state formulation posits that the universal wavefunction evolves unitarily forever; what appears to be a definite measurement outcome is a branch-relative fact — each observer in each branch experiences a definite outcome, but all outcomes exist simultaneously in the proliferating universal wavefunction. The many-worlds ontology requires positing a vast and unobservable multiplicity of simultaneously existing parallel realities. Phase Theory rejects branching entirely. By Proposition 6.1 and Lemma 6.1, only one globally admissible resolution exists for a completed measurement; the configurations corresponding to the "other branches" are globally inadmissible and do not exist. Moreover, the preferred basis problem is solved in Phase Theory by topological sector structure (Section 10), whereas Everett requires additional machinery (einselection, decoherence) to select the branch basis. Finally, the probability problem — deriving the Born rule in a branching picture — is non-trivial in Everett; Phase Theory derives it straightforwardly from admissibility measure (Theorem 6.2).

**Bohmian Mechanics.** Bohmian mechanics [11] posits that particles have definite positions at all times, guided by a pilot wave $\Psi$ satisfying the Schrödinger equation. Measurement outcomes are determined by the particle's actual position, which is a hidden variable — not accessible to quantum mechanical prediction but determining the outcome deterministically. Phase Theory is also deterministic (outcomes are determined by the global phase configuration), but it requires no pilot wave and no hidden positions. In Phase Theory, particles are not primitive objects but topological modes of the phase field (Paper 3 [43]); their positions are not hidden variables but emergent coordinates. The non-locality of the pilot wave — which must respond instantaneously to changes anywhere in the universe — is replaced in Phase Theory by the global character of admissibility, which is defined on the full composite space.

Both theories are non-local in emergent spacetime, but Phase Theory achieves this without introducing a new dynamical law (the guiding equation) beyond the admissibility constraint.

**QBism.** QBism [13] interprets the quantum state as an agent's degrees of belief (a subjective Bayesian probability assignment) and measurement outcomes as personal experiences of the agent. QBism is empirically indistinguishable from standard quantum mechanics in all tested regimes; it achieves this by making quantum mechanics a theory of agent behavior rather than physical reality. Phase Theory is diametrically opposed to QBism: it is fully realist and fully observer-independent. The quantum state (in the coherent limit) is not a description of an agent's beliefs but a coarse-grained description of a definite physical configuration Φ. The admissibility measure $\mu_\mathscr{I}$ is an objective measure on physical configuration space, not a subjective belief. Measurement outcomes are objective features of globally admissible phase configurations, not personal experiences. The success of QBism in making quantum mechanics epistemically coherent is reproduced by Phase Theory's admissibility measure, while Phase Theory goes further by providing an objective physical ontology.

Despite these ontological differences, Phase Theory reproduces the same measurement statistics as all three interpretations in all regimes where they themselves agree with standard quantum mechanics. The empirical differences are confined to regimes of extreme conditions (high mass interferometry, very precise Bell tests) where Phase Theory's novel predictions (Section 25) become relevant.

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## 22. Relation to Decoherence Programs and Einselection

Zurek's einselection program [8, 46] identifies "pointer states" as the preferred basis for measurement outcomes — those states of the system that are robust under environmental monitoring, in the sense that entanglement with the environment does not spread from pointer states to superpositions of pointer states. Einselection is a profound insight: it explains why certain macroscopic states (positions, roughly) are stable under environmental interaction while quantum superpositions thereof rapidly decohere. Phase Theory provides a deeper grounding for einselection: pointer states are topologically stable admissible configurations of the composite.

Formally, a pointer state $\Phi_{ptr}$ is characterized in Phase Theory by high admissibility margin: $\mathscr{I}[\Phi_{ptr} \oplus \Phi_e] \gg 0$ for typical environmental configurations $\Phi_e$. This means that pointer configurations remain admissible under small environmental perturbations — they are stable under the admissibility constraint. In contrast, superpositions of pointer states correspond to cross-term configurations that are generically inadmissible when coupled to the environment (Section 12). The einselection condition in Phase Theory is:

$$\partial\mathscr{I}[\Phi_{ptr} \oplus \Phi_e] / \partial\Phi_e \big|_{\Phi_e = \Phi_e^{(0)}} = 0 \quad \text{(to leading order)} \tag{22.1}$$

Equation (22.1) states that the admissibility functional is stationary with respect to small environmental perturbations at the pointer configuration — the pointer configuration is an admissibility fixed point under environmental coupling. This is the Phase Theory generalization of Zurek's condition that pointer states are stable under the Lindblad evolution generated by system–environment coupling. It is more general because it does not presuppose a Hilbert space structure or a Lindblad equation; it requires only the admissibility functional.

Einselection in Phase Theory is an approximation valid when the environmental coupling is weak — when the interaction terms $\mathscr{I}_{int}[\Phi_s, \Phi_e]$ are small compared to $\mathscr{I}[\Phi_s]$. In this weak-coupling limit, the admissibility stability condition (22.1) reduces to the standard einselection criterion. For strong coupling, Phase Theory provides the exact treatment: the pointer states are the admissibility-stable configurations under the full coupling $\mathscr{I}_{int}$, not just the weak-coupling approximation. This may be relevant for systems where the system-environment coupling is not weak — for instance, in the early-universe phase transitions that Phase Theory studies in Paper 7 of this series.

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## 23. Probability, Admissibility, and Formal Correspondence

We provide in this section the complete formal correspondence between quantum mechanical probability and Phase Theory admissibility measure, making precise the claim that probability in Phase Theory is objective, derived, and reduces to the Born measure in the coherent limit.

The admissibility measure $\mu_\mathscr{I}$ is defined as follows. Let $\mathscr{A} = \{\Phi \in \mathscr{M}_\Phi : \mathscr{I}[\Phi] \geq 0\}$ be the admissible region of phase configuration space. The admissibility measure of a measurable subset $S \subseteq \mathscr{M}_\Phi$ is:

$$\text{(23.1)} \quad \mu_\mathscr{I}(S) = \left[\int_{S \cap \mathscr{A}} e^{\mathscr{I}[\Phi]}\, d\mu_\mathscr{M}(\Phi)\right] / \left[\int_{\mathscr{A}} e^{\mathscr{I}[\Phi]}\, d\mu_\mathscr{M}(\Phi)\right]$$

where $d\mu_\mathscr{M}$ is the natural (Liouville-type) measure on $\mathscr{M}_\Phi$. The exponential weight $e^{\mathscr{I}[\Phi]}$ ensures that configurations with higher admissibility are weighted more strongly; configurations with $\mathscr{I}[\Phi] = 0$ (marginally admissible) receive weight 1; configurations with $\mathscr{I}[\Phi] < 0$ are excluded from $\mathscr{A}$ and receive zero weight.

We verify that $\mu_\mathscr{I}$ satisfies the Kolmogorov axioms of probability. **Axiom K1 (Non-negativity):** $\mu_\mathscr{I}(S) \geq 0$ for all S, since both numerator and denominator in (23.1) are integrals of non-negative functions. **Axiom K2 (Normalization):** $\mu_\mathscr{I}(\mathscr{M}_\Phi) = 1$, since $S = \mathscr{M}_\Phi$ gives numerator = denominator. **Axiom K3 (Countable additivity):** For disjoint measurable $S_1, S_2, \ldots$, $\mu_\mathscr{I}(\cup_i S_i) = \Sigma_i \mu_\mathscr{I}(S_i)$, which follows from the σ-additivity of the Lebesgue integral. Hence $\mu_\mathscr{I}$ is a probability measure on $\mathscr{M}_\Phi$. This is the formal content of Paper 5's Theorem 5.2 [45].

The coherent limit is characterized by the existence of a diffeomorphism $\iota: \mathscr{A}_{coh} \to CP^{K-1}$ from the coherent admissible sector of $\mathscr{M}_\Phi$ to the complex projective space $CP^{K-1}$, where K is the number of topological sectors accessible to the system. In this limit, the pullback of $\mu_\mathscr{I}$ under ι is the Fubini-Study measure $\omega_{FS}$ on $CP^{K-1}$, and the admissibility measure of topological sector $n_k$ becomes:

$$\text{(23.2)} \quad \mu_\mathscr{I}(\text{sector } n_k \mid \psi) = \int_{\pi_k^{-1}(\psi)} \omega_{FS} = |\langle n_k|\psi\rangle|^2 \equiv |\psi_k|^2$$

where $\pi_k: CP^{K-1} \to [0,1]$ is the projection onto the k-th coordinate modulus squared, and $|\psi\rangle$ is the coherent-limit quantum state. Equation (23.2) is the Born rule, derived as an identity of admissibility measures in the coherent limit.

**Corollary 6.2 (Gleason Recovery).**

The Fubini-Study measure $\omega_{FS}$ on $CP^{K-1}$ is the unique U(K)-invariant probability measure on $CP^{K-1}$ (by the uniqueness of Haar measure on compact groups). Since the admissibility functional $\mathscr{I}[\Phi]$ is invariant under all admissible phase reparametrizations — which in the coherent limit correspond to U(K) transformations — $\mu_{\mathscr{I}}$ restricted to the coherent sector is the unique admissibility-invariant probability measure, equal to $\omega_{FS}$. This implies Gleason's theorem [24]: the unique non-contextual probability assignment on the lattice of Hilbert subspaces of dimension ≥ 3 is the Born measure. Gleason's theorem is thus a corollary of admissibility measure uniqueness.

Outside the coherent limit — in regimes where phase configurations are not well-described by complex amplitudes, such as very high energy density, early-universe phases, or systems near topological phase transitions — equation (23.2) no longer holds exactly. Phase Theory predicts deviations from the Born rule of order $\delta P \sim (\mathscr{I}_{non\text{-}coh} / \mathscr{I}_{coh})^2$, where $\mathscr{I}_{non\text{-}coh}$ is the non-coherent contribution to the admissibility functional. In laboratory quantum mechanics, $\mathscr{I}_{non\text{-}coh} / \mathscr{I}_{coh} \sim 10^{-8}$ or smaller, yielding Born-rule deviations of order $10^{-16}$ — far below any current measurement sensitivity. These are among the predictions listed in Section 25.

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## 24. Experimental Consistency with Known Measurement Phenomena

Phase Theory is required to be consistent with all known measurement phenomena. We verify this systematically, covering the principal empirical phenomena that any theory of measurement must explain.

**(1) Discrete outcomes.** Every quantum measurement produces one of a discrete set of outcomes, corresponding to the eigenvalues of the measured observable. Phase Theory derives this from topological winding numbers (Paper 3 [43], Section

10): stable admissible configurations are in definite topological sectors characterized by integer winding numbers. Outcome discreteness is a consequence of topological quantization, not postulated. ✓

**(2) Born-rule statistics.** Repeated measurements on identically prepared systems produce outcome frequencies converging to the Born probabilities $P(n_k) = |\psi_k|^2$. Phase Theory derives these from the admissibility measure $\mu_{\mathscr{I}}$ in the coherent limit (Theorem 6.2, Section 13). For any test of Born-rule statistics within current experimental precision, Phase Theory predicts deviations of order $10^{-16}$ or smaller — far below detection threshold. ✓

**(3) Decoherence timescales.** The timescale on which quantum superpositions of macroscopic states decohere is given by $\tau_{dec} \sim \tau_{relax} / (\Delta x/x_{th})^2$, measured in experiments such as Brune et al. [29] and Hackermüller et al. [28]. Phase Theory reproduces these timescales as admissibility pruning timescales (Section 12, equation 12.2): $\tau_{prune} = \tau_{dec}$. The quantitative agreement follows from the mapping between the environmental coupling in standard QM and the environmental admissibility interaction terms $\mathscr{I}_{int}[\Phi_s, \Phi_e]$. ✓

**(4) Contextuality.** Measurement statistics depend on the broader experimental context (Kochen-Specker [14], Bell [12], Aspect et al. [15]). Phase Theory derives contextuality from global admissibility non-separability (Corollary 6.1, Section 15). All quantitative predictions — violation magnitudes, state-space dependencies — are reproduced in the coherent limit by the admissibility measure reducing to standard quantum mechanics. ✓

**(5) Interference and its loss.** Quantum systems exhibit interference when no which-path information is present, and lose interference when which-path information is recorded (Arndt et al. [27], Young's double slit). Phase Theory explains this as the presence or absence of admissible cross-routing configurations (Section 12): interference configurations are admissible when no which-path register is coupled; they become inadmissible when the register is coupled and correlated (equation 12.1). ✓

**(6) Spin measurement (Stern-Gerlach).** A spin-1/2 particle entering a Stern-Gerlach device is deflected either up or down, never continuously. The deflection angle is discrete and corresponds to the magnetic quantum number. Phase Theory accounts for this as admissibility selection between the two winding sectors of the spin phase configuration ($n = \pm\frac{1}{2}$) combined with the apparatus coupling to the magnetic field phase (emergent from phase curvature, Paper 4 [44]). The discrete deflection corresponds to the discrete winding sectors; continuous deflections would require non-integer winding numbers, which are topologically forbidden. ✓

**(7) Photon polarization measurement.** A photon polarization measurement along any axis gives a binary outcome (H or V, or + or −). Phase Theory represents photon polarization as the orientation of a propagating phase excitation mode (Paper 3 [43]). The measurement selects the admissible excitation orientation consistent with the apparatus (polarizer) phase configuration. Binary outcomes correspond to the two admissible orientations for a given polarizer axis; intermediate orientations are inadmissible for the measurement composite. ✓

**(8) Double-slit with which-path detection.** When a which-path detector is placed at one slit, the interference pattern disappears. Kim et al. [26] and many other groups have verified this in multiple physical systems. Phase Theory accounts for this as admissibility elimination of cross-routing configurations upon coupling with the path detector (Section 12, 19). The which-path detector's admissibility constraints make cross-slit routing configurations globally inadmissible, leaving only the single-slit routing — no interference. ✓

We conclude that no currently observed measurement phenomenon requires any extension of Phase Theory. All eight classes of phenomena are reproduced within the existing framework of the admissibility functional, topological discreteness, and admissibility measure, without additional postulates.

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## 25. Novel Predictions and Falsifiable Deviations

A theory is scientifically meaningful only insofar as it makes predictions distinguishable from its competitors. Phase Theory is empirically equivalent to standard quantum mechanics in all currently tested regimes, but it makes six classes of novel, falsifiable predictions in regimes accessible to near-future experiment. We enumerate these in detail, with proposed experimental protocols for each class.

**Prediction Class 1: Topology-dependent Born-rule deviations.** In the coherent limit, Phase Theory reproduces the Born rule exactly (Theorem 6.2). Outside the coherent limit — in systems with highly non-trivial phase topology, such as those near topological phase transitions, in exotic phases of matter, or under extreme confinement — Phase Theory predicts Born-rule deviations of order $\delta P \sim 10^{-8}$ in carefully controlled experiments. The predicted signature is a systematic deviation in outcome frequencies from $|\psi_k|^2$ in a direction determined by the topological charge of the system's phase configuration, not by random noise. Specifically, the correction is:

$$P(n_k) = |\psi_k|^2 + \varepsilon_k \cdot \delta_{topo} + O(\delta_{topo}^2) \quad (25.1)$$

where $\varepsilon_k$ are O(1) coefficients with $\Sigma_k \varepsilon_k = 0$ (total probability conserved), and $\delta_{topo}$ is the topological deviation parameter characterizing the non-coherent sector contribution. This prediction is in principle testable in topological quantum matter systems (topological insulators, fractional quantum Hall states). Experimental protocol: measure Born-rule statistics for a topological qubit in a system near a topological phase transition, comparing with the same qubit far from the transition. The predicted deviation $\delta P \sim 10^{-8}$ requires at least $10^{16}$ repeated measurements at $10^{-8}$ relative precision — challenging but within the range of future quantum metrological technology.

**Prediction Class 2: Admissibility-constrained interference revival.** Under specific apparatus phase configurations — in particular, apparatus configurations with unusually high internal coherence (low thermal entropy, near ground state) — Phase Theory predicts that environmental decoherence can be partially reversed without quantum erasure. The mechanism: when the apparatus's constraint density $\rho_c[\Phi_a]$ is anomalously low (e.g., for a nearly pure apparatus state), the admissibility pruning of cross-term configurations is incomplete, and partial revival of interference is predicted on a timescale:

$$\tau_{revival} \sim (\tau_{prune} \cdot \tau_{coherence})^{1/2} \cdot \exp(+\rho_c^{-1} \cdot N_a) \quad (25.2)$$

Experimental protocol: interferometry of a quantum system coupled to a near-ground-state mechanical resonator (membrane-in-the-middle or levitated nanoparticle setup). The predicted revival should appear as partial restoration of interference visibility without explicit erasure of which-path information.

**Prediction Class 3: Macroscopic coherence N-dependence.** Phase Theory predicts a specific power-law dependence of the decoherence suppression timescale on system size N: $\tau_{prune} \sim N^{-\gamma}$, where $\gamma$ is a dimensionless exponent determined by the admissibility functional for the specific system. For typical molecular systems, $\gamma \approx 2/3$ (from the scaling of the topological barrier height with N). This differs from CSL's prediction (which scales as mass-squared density, with a different functional form) and from standard decoherence (which scales as $\Delta x^2$ in the Caldeira-Leggett model). The N-dependent decoherence signature is testable in macromolecule interferometry (Arndt et al. [27], Hackermüller et al. [28] for $C_{60}$ and $C_{70}$ fullerenes; Eibenberger et al. for larger molecules). Future experiments with molecules of $N \sim 10^3$–$10^4$ atoms should be able to distinguish $\gamma = 2/3$ (Phase Theory) from the CSL exponent.

**Prediction Class 4: Dark-sector recoil in measurement.** Phase Theory, developed in Papers 3–4 of this series [43, 44], predicts the existence of a dark sector — topological phase configurations that couple to standard-model configurations only through gravitational-scale admissibility interactions. For particles with dark-sector interactions, measurement produces not only the standard-model recoil but also a dark-sector admissibility adjustment. This manifests as a small but systematic deviation in the recoil momentum distribution measured in precision atomic physics experiments: the recoil distribution develops a non-Gaussian tail at the level $\delta P_{recoil} \sim \varepsilon_{dark} \sim 10^{-6}$, where $\varepsilon_{dark}$ is the dark-sector coupling parameter derivable from the admissibility functional. Experimental protocol: precision recoil spectroscopy of cesium or rubidium atoms in coherent scattering experiments, looking for systematic non-Gaussian deviations in momentum transfer distributions.

**Prediction Class 5: Wigner's Friend inconsistency threshold.** Phase Theory predicts a critical system size $N_c$ above which Wigner's Friend protocols become experimentally distinguishable from Many-Worlds predictions. In MWI, a Wigner-type measurement of the composite (System + Friend) can in principle reveal the coherence of the post-Friend-measurement state. In Phase Theory, no such coherence survives once the Friend's apparatus has satisfied $\rho_c[\Phi_F] \geq \rho_*$, because admissibility pruning has already selected a definite outcome. The critical size $N_c \sim \rho_* / \alpha' \sim 10^6$–$10^8$ atoms (a mesoscopic "Friend" apparatus). For systems smaller than $N_c$, partial coherence survives and both Phase Theory and MWI predict some Wigner-type interference. For $N > N_c$, Phase Theory predicts complete decoherence and no Wigner interference; MWI predicts (in principle) recoverable coherence. Near-future quantum computing experiments — using arrays of entangled qubits as the "Friend" — can probe this boundary.

**Prediction Class 6: Phase-topology signature in Bell tests.** Phase Theory predicts that specific correlator combinations beyond the standard CHSH should show deviations from the Tsirelson bound $2\sqrt{2}$ at high experimental precision. These deviations arise from the non-trivial topology of $\mathscr{M}_\Phi$: for specific entangled states near topological boundaries in phase configuration space, the Tsirelson bound is not exactly achieved because the maximizing configurations are topologically obstructed. The predicted deviation is of order $\delta\langle CHSH\rangle \sim \varepsilon_{topo} \sim 10^{-6}$ below $2\sqrt{2}$, direction-dependent in the measurement-angle space. Experimental protocol: high-precision loophole-free Bell tests with angular resolution $10^{-4}$ radians and statistical precision below $10^{-6}$ in the correlator magnitude. Future experiments using photon pairs from space-based entanglement sources (following the Micius satellite program) may reach the required precision.

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## 26. Implications for the Quantum-Classical Divide

The quantum-classical divide — the question of why macroscopic objects behave according to classical deterministic dynamics while microscopic systems exhibit characteristically quantum phenomena — is one of the deepest questions in the foundations of physics. The standard answer, via decoherence, is that environmental entanglement suppresses quantum interference on very short timescales for macroscopic systems, effectively rendering them classical. While decoherence explains the empirical absence of observable quantum superpositions in macroscopic systems, it does not explain why classical dynamics are exactly obeyed by these systems, or why there is a sharp (even if gradual) transition from quantum to classical behavior. Phase Theory provides a more fundamental answer.

**Theorem 6.4 (Classicality as High-Constraint-Density Limit).**

In the limit $\rho_c[\Phi] \to \infty$ (infinite constraint density), the admissibility-allowed dynamics of any phase structure converge to classical deterministic dynamics. More precisely: let $\Phi_{macro}$ be a macroscopic phase configuration with constraint density $\rho_c[\Phi_{macro}] = \rho >> 1$. Then the admissibility-allowed trajectories $\Phi_{macro}(\lambda)$ in $\mathscr{M}_\Phi$ converge, as $\rho \to \infty$, to trajectories satisfying the classical equations of motion of the corresponding emergent effective theory (Paper 2 [42]), with corrections of order $\rho^{-1}$.

***Proof sketch.*** *A macroscopic phase configuration with constraint density $\rho >> 1$ has $O(\rho \cdot Vol_{\mathscr{M}}(\text{supp } \Phi_{macro}))$ independent admissibility constraints active. Each constraint restricts the allowed trajectory in $\mathscr{M}_\Phi$. In the limit $\rho \to \infty$, the constraints are so numerous and dense that the allowed trajectory is determined up to $O(\rho^{-1})$ deviations — the trajectory is confined to the minimum of the admissibility functional, which by the variational principle of Paper 2 [42] corresponds to the classical equation of motion. Quantum fluctuations in Phase Theory correspond to $O(\rho^{-1})$ deviations from the classical trajectory, consistent with the semiclassical ħ-expansion of standard quantum mechanics (ħ/action ~ 1/ρ in appropriate units).* ∎

Theorem 6.4 establishes that classicality is not an approximation imposed on quantum mechanics from outside, but a derived limit of Phase Theory. As the constraint density of a system increases — as it becomes more macroscopic, more thermalized, more entangled with its environment — its admissibility-allowed dynamics become progressively more classical. The transition from quantum to classical behavior is therefore not a sharp divide but a gradient in constraint density: small ρ corresponds to quantum behavior, large ρ corresponds to classical behavior, and intermediate ρ corresponds to the mesoscopic regime where both quantum and classical features coexist.

This framework resolves several specific puzzles about the quantum-classical transition. First, the "Schrödinger's cat" problem: why do macroscopic objects (cats) never appear in quantum superpositions? Because their constraint density $\rho_c \sim 10^{23}$ makes the admissibility-allowed configuration space entirely classical — all quantum-looking configurations are inadmissible (equation 7.3). Second, the "pointer basis" problem: which basis of the quantum state space corresponds to classical configurations? The pointer basis corresponds to the topologically stable admissible configurations at high constraint density — those satisfying the einselection criterion (22.1) — which are exactly the configurations described by classical mechanics. Third, the "emergence of space" question: why does classical mechanics operate in three-dimensional position space when Phase Theory's fundamental space is $\mathscr{M}_\Phi$? Because position space is the emergent effective description of the high-ρ limit of admissibility dynamics (Paper 2 [42]), valid when the phase configuration is slowly varying and coherent on the emergent spatial scale.

The quantum-classical gradient picture has a direct experimental implication. As systems are made progressively larger — from atoms to molecules to nanoparticles to mesoscopic objects — Phase Theory predicts a specific functional form for the evolution of quantum features (interference visibility, entanglement entropy, Bell violation magnitude) with system size. This functional form is governed by $\rho_c(N)$ and is quantitatively different from both CSL and standard decoherence predictions in the intermediate regime $N \sim 10^3$–$10^8$. This is the same regime probed by Prediction Class 3 (Section 25), making it a particularly fertile testing ground for Phase Theory's account of the quantum-classical transition.

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## 27. Implications for Quantum Information and Computation

Phase Theory has direct and significant implications for quantum information science. The central concepts of quantum information — entanglement, decoherence, quantum error correction, channel capacity — acquire new interpretations within the admissibility framework, and at least one new quantum information result is derivable within Phase Theory.

In Phase Theory, **quantum entanglement** is non-separability of the admissibility functional (Proposition 6.0, equation 6.5). The entanglement entropy of a bipartite system $S = A \cup B$ is, in Phase Theory, the admissibility entropy $H_\mathscr{I}(A:B)$ — the logarithm of the volume of admissible joint configurations of A and B relative to the product of their marginal admissible configuration spaces:

$$H_\mathscr{I}(A:B) = -\log \left[\mathrm{Vol}_\mathscr{M}(\mathscr{A}_{AB}) / (\mathrm{Vol}_\mathscr{M}(\mathscr{A}_A) \cdot \mathrm{Vol}_\mathscr{M}(\mathscr{A}_B))\right] \tag{27.1}$$

In the coherent limit, $H_\mathscr{I}(A:B)$ reduces to the standard von Neumann entanglement entropy $S(\rho_A) = -\mathrm{Tr}(\rho_A \log \rho_A)$, by the correspondence between admissibility measure and the Fubini-Study measure established in Section 23. Outside the coherent limit, $H_\mathscr{I}(A:B)$ provides a generalized entanglement measure valid in regimes where the von Neumann entropy is not well-defined.

**Quantum error correction** in Phase Theory is a protocol for maintaining global admissibility of a composite phase configuration against local perturbations. Decoherence in quantum computers corresponds to admissibility pruning of computational phase configurations by environmental coupling: errors are the result of environmental constraints eliminating valid computational configurations. Quantum error correcting codes are subspaces of the computational configuration space that are robust under admissibility perturbations — that is, the code subspace has a high admissibility margin against the class of errors considered, satisfying $\mathscr{I}[\Phi_{code} \oplus \Phi_{error} \oplus \Phi_{environment}] \geq 0$ for all errors in the error class. Extending coherence times in quantum computers requires increasing the admissibility margin of the code configurations — a principle that may suggest new code constructions in the topological regime.

The **holographic principle** — that the information content of a spatial region is bounded by its boundary area — acquires a natural interpretation in Phase Theory. In Phase Theory, the admissibility constraints on a spatial region $\Omega$ are determined by boundary data on $\partial\Omega$ (the boundary of the region in emergent spacetime). This is the Phase Theory counterpart of holography: the bulk admissibility is determined by boundary admissibility constraints. Equation (27.1) can be applied to the holographic case, with A = bulk and B = boundary, giving a holographic area bound that in the coherent limit recovers the Bekenstein-Hawking entropy bound.

A new quantum information result derivable within Phase Theory is the following. **Proposition (Admissibility Channel Capacity):** The quantum channel capacity of a physical channel modeled as a phase coupling $\Phi_{in} \oplus \Phi_{channel} \oplus \Phi_{noise}$ is bounded above by the admissibility entropy of the channel phase configuration: $C \leq H_{\mathscr{I}}(in:out)$. In the coherent limit, this reduces to the Holevo bound. Outside the coherent limit, Phase Theory predicts corrections to the Holevo bound proportional to the non-coherent admissibility contribution, which may be measurable in extremely noisy or topologically complex channels.

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## 28. Philosophical Implications — Ontology Without Observers

Phase Theory has profound philosophical consequences for the interpretation of quantum mechanics and for the philosophy of physics more broadly. We address five philosophical themes: realism, observer-independence, determinism, non-locality, and structural realism.

**Realism.** Phase Theory is strongly realist: there is a definite phase configuration $\Phi \in \mathscr{M}_\Phi$ at all times, constituting the complete physical reality. There is no superposition of realities, no parallel branches, no indefinite physical properties. Every physical observable is determined — in principle — by the global phase configuration, even though many observables may be inaccessible to subsystems of the composite (epistemic limitations). This vindicates Einstein's famous complaint that "God does not play dice" in the following sense: the apparent randomness of quantum outcomes is not fundamental indeterminism in the physical world but epistemic uncertainty about the global phase configuration. Einstein's realist intuitions were correct; his specific proposals (local hidden variables) were wrong, but Phase Theory shows that a different kind of deeper reality — non-local in emergent spacetime but governed by global admissibility — is available.

**Observer-independence.** Physical reality in Phase Theory does not depend on observation, measurement, or consciousness. The global phase configuration $\Phi$ exists and is governed by $\mathscr{I}[\Phi] \geq 0$ regardless of whether any observer is present, has access, or performs measurements. This is the strongest possible version of observer-independence: not merely that observers do not disturb physical reality (beyond their physical effect as macroscopic phase structures), but that the concept of "observer" has no special ontological role. This eliminates what Einstein called "the spooky epistemology" of the Copenhagen interpretation — the doctrine that quantum mechanics is about measurement results rather than physical reality.

**Determinism.** Phase evolution in Phase Theory is deterministic: given the initial global phase configuration $\Phi(0) \in \mathscr{M}_\Phi$, the subsequent admissibility-constrained trajectory $\Phi(\lambda)$ is uniquely determined (by Axioms I–IV and the admissibility functional). This is a stronger form of determinism than classical mechanics, because it requires no specification of a law of motion beyond admissibility: all admissible trajectories are realized, and all inadmissible ones are forbidden. Apparent randomness — the Born-rule statistics of measurement outcomes — arises from epistemic limitations: subsystems of the composite cannot access the global phase configuration, and their predictions are therefore probabilistic descriptions of an objective, determined reality.

**Non-locality and the locality paradox.** Phase Theory is non-local in emergent spacetime: the admissibility constraint $\mathscr{I}[\Phi_{total}] \geq 0$ correlates spatially separated configurations. But this non-locality does not violate any physical locality principle, because it occurs in phase configuration space $\mathscr{M}_\Phi$ — the fundamental ontological substrate — rather than in emergent spacetime. The no-signaling theorem (Section 18) ensures that admissibility correlations cannot be used to transmit information superluminally in emergent spacetime. The locality paradox of quantum mechanics — how can spatially separated particles be correlated without communication? — is dissolved: the correlations arise from global admissibility constraints in $\mathscr{M}_\Phi$, a space in which "spatial separation" in emergent coordinates is not the relevant notion of distance.

**Structural realism.** Phase Theory is naturally aligned with ontic structural realism in the philosophy of physics — the view that physical reality consists of structures (relations, symmetries, constraints) rather than individual objects with intrinsic properties. In Phase Theory, there are no primitive objects: particles, fields, and spacetime are emergent structures. What is primitive is the phase configuration and its admissibility — a structural, relational, global entity. The objects of physics (particles, fields) are patterns in this structure, not independent beings. This vindicates the structural realist program of French and Ladyman [see 46], who argue that quantum mechanics requires giving up object-oriented ontology in favor of structure. Phase Theory provides the explicit structural ontology that structural realism calls for.

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## 29. Relation to Gravitational Decoherence

Gravitational decoherence — the hypothesis that gravity plays a fundamental role in the collapse of quantum superpositions — has been proposed by Penrose [36, 37] and independently by Diósi [34]. In the Penrose scheme, a quantum superposition of two spatial configurations of a massive body generates a superposition of two spacetime geometries; the gravitational self-energy of this superposition EG = E_Penrose exceeds $\hbar/\tau$ for $\tau$ equal to the collapse timescale, leading to spontaneous collapse. Diósi's model implements this via a stochastic noise term in the Schrödinger equation, proportional to the Newtonian gravitational potential. Phase Theory provides a unifying reinterpretation of both proposals within the admissibility framework.

In Phase Theory, gravity is an emergent effective description of phase curvature — the curvature of the phase field $\Phi$ in $\mathscr{M}\_\Phi$, which manifests as spacetime curvature in the emergent geometric description of Paper 2 [42]. Gravitational decoherence therefore corresponds, in Phase Theory, to admissibility pruning mediated by the gravitational sector of the phase field: the gravitational phase configuration $\Phi$_grav introduces admissibility constraints on the matter phase configuration $\Phi$_matter, enforcing that configurations with large gravitational self-energy are admissibility-suppressed.

Formally, the Penrose collapse criterion EG ≥ ℏ/τ corresponds in Phase Theory to the condition that the gravitational interaction term 𝒥_int[Φ_matter, Φ_grav] drops below the admissibility threshold:

$$\mathscr{I}\_int[\Phi\_matter^{super}, \Phi\_grav] < -\mathscr{I}\_threshold^{grav} \iff E\_G \geq \hbar / \tau\_prune \quad (29.1)$$

This establishes that the Penrose criterion is an approximation to the admissibility pruning condition for the gravitational sector. Phase Theory unifies gravitational decoherence (Penrose-Diósi) with environmental decoherence (Zurek-Joos) under the single admissibility framework: both are instances of admissibility pruning, differing only in which sector of Φ_total dominates the pruning (gravitational vs. electromagnetic vs. phonon, depending on the system). No separate "quantum gravity" mechanism is needed for collapse; gravity contributes to admissibility constraints as one sector among others, albeit a sector that becomes dominant for sufficiently massive superpositions (M ≥ M_Planck / N^{1/2} for N-particle composites).

The Diósi-Penrose collapse rate (τ_DP = ℏ/EG) corresponds, within Phase Theory, to the admissibility pruning timescale in the gravitational sector. The agreement of Phase Theory's prediction with the Diósi-Penrose rate is a non-trivial consistency check: it requires that the gravitational admissibility threshold 𝒥_threshold^{grav} equals EG/ℏ in the appropriate units, a condition that follows from the emergent general relativity structure of Paper 2 [42] and the identification of the gravitational energy with the phase curvature action.

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## 30. Conclusion

This paper has developed a complete, rigorous treatment of measurement within Phase Theory, demonstrating that the measurement problem — one of the deepest and most persistent conceptual difficulties in the foundations of physics — is dissolved rather than solved once the correct ontological primitives are identified. The central result is the following.

**Measurement is deterministic phase constraint resolution under global admissibility.** It is not a primitive physical process, not an exception to the dynamics of the theory, not a process requiring a special postulate, and not a process requiring an observer. It is an ordinary physical interaction between system, apparatus, and environment, governed by the admissibility functional $\mathscr{I}[\Phi\_total] \geq 0$, distinguished from other interactions only by the macroscopic character of the apparatus (Constraint Density $\rho\_c[\Phi\_a] \geq \rho\_*$), which generates the admissibility pruning power sufficient to select unique, definite outcomes.

The five principal conceptual dissolutions achieved in this paper are: **(1) Collapse is dissolved** — what appears as wavefunction collapse is admissibility pruning, a continuous deterministic process that eliminates globally inadmissible configurations without any primitive jump (Proposition 6.1, Theorem 6.1). **(2) Observer privilege is dissolved** — observers are macroscopic phase structures subject to the same admissibility rules as all other systems; consciousness plays no causal role in outcome selection (Proposition 6.3, Section 14). **(3) The quantum-classical divide is dissolved** — classicality is the high-constraint-density limit of admissibility dynamics, a derived regime rather than a separately imposed approximation (Theorem 6.4, Section 26). **(4) The preferred basis problem is dissolved** — the outcome basis is determined by the topological sector structure of $\mathscr{M}\_\Phi$, which is independent of any choice of Hilbert space basis (Section 10). **(5) Contextuality is explained** — measurement context-dependence is a structural consequence of the global non-separability of the admissibility functional, requiring no additional mystery or paradox (Corollary 6.1, Section 15).

In addition, Phase Theory has been shown to reproduce all known quantum mechanical predictions for measurement in tested regimes (Section 24), to offer novel falsifiable predictions distinguishing it from standard quantum mechanics and from collapse theories (Section 25), and to provide a unified framework subsuming decoherence, einselection, and gravitational collapse under the single admissibility principle (Sections 12, 22, 29).

Phase Theory VI completes the treatment of quantum foundations within this series. Papers 7–10 will apply the Phase Theory framework to cosmology (the origin of the universe as an admissibility initial condition), thermodynamics (entropy as admissibility measure), dark matter and dark energy (dark-sector topological configurations), and experimental predictions across all sectors. The broader program aims to show that once phase is taken as the sole ontological primitive, and global admissibility as the sole physical law, the full structure of physical reality — spacetime, matter, forces, probability, and classicality — emerges without any additional postulates.

The conclusion is simple and far-reaching: once phase is primary, the measurement problem disappears — not because it has been solved within the old framework, but because the framework that generated it no longer applies.

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## Notes

[1] The term "dissolution" is used deliberately to distinguish Phase Theory's resolution from prior "solutions" that operate within the existing conceptual framework. A solution to the measurement problem would explain, within standard quantum mechanics and its ontological primitives, why collapse occurs and what determines its outcomes. A dissolution shows that the problem cannot even be stated in the correct ontological framework. This distinction is philosophically significant and follows Wittgenstein's distinction between solving a problem and dissolving it by showing it was not well-posed.

[2] The notation ⊕ for phase composition is chosen to distinguish it clearly from tensor product ⊗ (Hilbert space) and direct sum ⊕ (vector spaces). In Phase Theory, ⊕ is a topological gluing operation defined on $\mathscr{M}$_Φ; its precise definition is given in Paper 1 [41], Section 4. It is associative but not commutative in general, as the gluing depends on the ordering of boundary identifications.

[3] The independence assumption for apparatus constraints (Section 7) is the Phase Theory analog of the "independence of records" assumption in decoherence theory (Zurek [8]). Both assume that the many degrees of freedom of the apparatus are approximately uncorrelated in their constraint structure, which holds for thermal (high-temperature) macroscopic systems. For highly ordered macroscopic systems (crystals, lasers), the effective number of independent constraints is reduced, and correspondingly the admissibility pruning power is reduced — consistent with the known fact that highly ordered systems can exhibit quantum coherence at larger scales.

[4] The Tsirelson bound $2\sqrt{2}$ was first derived by Tsirelson [25] as the maximum quantum violation of the CHSH inequality. The Phase Theory derivation in Theorem 6.3 provides a new physical interpretation: the bound is a maximum of admissible global phase correlations, not merely a mathematical consequence of Hilbert space structure. This interpretation may yield new insights into why nature respects the Tsirelson bound rather than the algebraic maximum of 4 or the Popescu-Rohrlich bound of 4.

[5] The philosophical position of Phase Theory — deterministic, realist, observer-independent, structuralist — is close to what Einstein hoped to find in a completion

of quantum mechanics. The key difference from Einstein's expectations is that Phase Theory achieves realism and determinism at the cost of local separability: the admissibility functional is genuinely global, and the correlations between spatially separated events are built into the structure of $\mathscr{M}_\Phi$ rather than transmitted by any local mechanism. Bell's theorem [12] showed that local hidden variables cannot reproduce quantum correlations; Phase Theory avoids this by providing non-local (global admissibility) rather than local structure.

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